

Quantum geometry of collective pairing fluctuations in the superfluid weight of multiband superconductors

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Beyond the frozen-pairing contribution, the superfluid weight of a multiband superconductor contains a correction from the self-consistent relaxation of the pairing amplitudes under a phase twist. We show that this correction has a quantum-geometric origin, not in the space of Bloch states but in the space of collective pairing fields. For an attractive multiband Hubbard model at the mean-field level, the thermodynamic Hessian governing the pairing response coincides with the static Gaussian pair-fluctuation kernel at zero momentum at every temperature below T_c . Since gauge invariance relates a phase twist to a finite center-of-mass momentum of the pair field, the superfluid weight is set by the small-momentum curvature of the eigenvalue branch that evolves from the Goldstone mode, and the relaxation correction is a stiffness-weighted quantum metric of the corresponding soft eigenvector on the manifold of pairing configurations. The same construction yields the Ginzburg-Landau gradient coefficient with the identical geometric correction, hence the coherence-length tensor, and, with the low-frequency dynamics of the kernel, the pair-mass tensor. Lieb-lattice calculations verify the equivalence of the thermodynamic, fluctuation-kernel, and soft-mode formulations and the independence of the superfluid weight from the orbital embedding.

I. INTRODUCTION

The superfluid weight [1, 2] of a multiband superconductor is determined not only by the dispersion of its energy bands but also by the quantum geometry of their Bloch states [3, 4]. In particular, interband processes can generate a geometric contribution that remains finite even in flat-band systems, where the conventional contribution associated with band dispersion is suppressed. This connection between quantum geometry and superfluidity has motivated extensive interest in multiband and flat-band superconductors [5–15]. Experiments on moiré graphene support this picture: the measured superfluid stiffness of magic-angle twisted bilayer graphene exceeds the conventional band-dispersion estimate by an order of magnitude, consistent with a dominant quantum-geometric contribution [16, 17]. Its temperature and current dependence point to a nodal gap [17, 18], while short coherence lengths comparable to the interparticle distance and gap-to- T_c ratios far above the BCS value place these systems in the strong-coupling regime [19–22].

The geometry of the underlying Bloch states, however, is not the only source of corrections to the superfluid weight. In a multiorbital superconductor the pairing amplitudes on different orbitals are independent degrees of freedom, so a phase twist can change their magnitudes and relative phases even when the equilibrium pairing is spatially uniform; a superfluid weight computed with the amplitudes held fixed misses this response [23–27]. Huhtinen *et al.* [24] identified the resulting relaxation correction and expressed it through the order-parameter response and the Hessian of the mean-field thermodynamic potential, which provides a general formulation of the self-consistent superfluid weight. Less explicit in that formulation is its connection to collective pairing fluctuations and to the momentum dependence of their

eigenmodes [28–30], even though a phase twist couples directly to the finite-momentum pair field while the thermodynamic Hessian describes only its static response at zero momentum.

In this work, we connect these two descriptions directly. For an attractive multiband Hubbard model at the mean-field saddle-point level, we show that the Hessian of the thermodynamic potential with respect to the complex pairing fields coincides with the Gaussian pair-fluctuation kernel at zero momentum and frequency, at every temperature below T_c . Since gauge invariance maps a uniform phase twist onto a finite center-of-mass momentum $\mathbf{Q}$ of the pair field, the self-consistent superfluid weight follows from the small-momentum curvature of the eigenvalue branch of this kernel that evolves from the Goldstone mode. The relaxation correction arises because at finite $\mathbf{Q}$ the Goldstone mode couples to the massive amplitude and relative-phase channels; integrating them out yields a Schur complement whose subtracted term is a stiffness-weighted quantum metric of the soft eigenvector on the manifold of pairing configurations. This collective metric is distinct from the Bloch quantum metric, and it vanishes when the Goldstone mode decouples to linear order in $\mathbf{Q}$. At T_c the same projection onto the critical pairing channel gives the Ginzburg-Landau (GL) gradient coefficient with the same geometric correction, and hence the coherence-length tensor [28], whereas the pair mass additionally requires the low-frequency dynamics of the kernel. For the Lieb lattice we verify numerically that the thermodynamic, fluctuation-kernel, and soft-mode formulations agree, that the frozen-pairing and relaxation contributions are separately embedding dependent while their sum is not, and that in the disconnected limit the relaxation correction cancels the spurious frozen-pairing stiffness [24].

The paper is organized as follows. In Sec. II, we develop the thermodynamic formulation of the superfluid

weight and isolate the pairing-relaxation correction. In Sec. III, we connect it to Gaussian pair fluctuations and develop the soft-mode and quantum-geometric description. In Sec. IV, we assemble the resulting superfluid weight and discuss the conditions under which the relaxation correction vanishes. In Sec. V, we extend the construction to the critical pairing channel and derive the coherence-length and pair-mass tensors. In Sec. VI, we summarize the assumptions and scope of the approach, and in Sec. VII, we present the Lieb-lattice benchmark. In Sec. VIII, we conclude the paper. The appendices derive the kernel-Hessian relation in an alternative way, examine the kernel under a change of orbital embedding, and establish the gauge relations between phase-twist and pair-momentum derivatives.

II. THERMODYNAMIC FORMULATION

We consider an attractive Hubbard model on a translationally invariant lattice with N_S sublattice sites (orbitals) per unit cell,

$$H = \sum_{ll'} \sum_{SS'\sigma} t_{SS'}^\sigma(\mathbf{R}_l - \mathbf{R}_{l'}) c_{lS\sigma}^\dagger c_{l'S'\sigma} - \mu \sum_{lS\sigma} n_{lS\sigma} - U \sum_{lS} n_{lS\uparrow} n_{lS\downarrow}, \quad U > 0, \quad (1)$$

where $c_{lS\sigma}^\dagger$ and $c_{lS\sigma}$ create and annihilate a fermion of spin σ on orbital S of unit cell l , and $n_{lS\sigma} = c_{lS\sigma}^\dagger c_{lS\sigma}$. The index $l = 1, \dots, N_c$ labels the unit cells, located at $\mathbf{R}_l$, and $S = 1, \dots, N_S$ labels the orbitals within a unit cell; the indices i and j are reserved throughout for Cartesian directions, and the imaginary unit is written in roman type as i . We denote the position of orbital S relative to the unit-cell origin by $\mathbf{x}_S$, so that its real-space position is $\mathbf{r}_{lS} = \mathbf{R}_l + \mathbf{x}_S$. Translational invariance implies that the hopping amplitudes depend only on the unit-cell displacement $\mathbf{R} = \mathbf{R}_l - \mathbf{R}_{l'}$, while Hermiticity requires $t_{SS'}^\sigma(\mathbf{R}) = [t_{S'S}^\sigma(-\mathbf{R})]^*$. The last term in Eq. (1) describes an on-site attraction of strength U between opposite-spin fermions, taken for simplicity to be identical on every orbital. Throughout, we set $\hbar = k_B = 1$ and work in the grand-canonical ensemble at fixed chemical potential μ .

We use the Fourier transformation $c_{lS\sigma} = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_{lS}} c_{\mathbf{k}S\sigma}$, where the orbital positions appear explicitly, and $\mathbf{k}$ runs over the first Brillouin zone (BZ) where $\sum_{\mathbf{k} \in \text{BZ}} 1 = N_c$. The single-particle part of Eq. (1) is then described by the $N_S \times N_S$ Bloch matrix $[h_{\mathbf{k}\sigma}]_{SS'} = \sum_{\mathbf{R}} t_{SS'}^\sigma(\mathbf{R}) e^{-i\mathbf{k} \cdot (\mathbf{R} + \mathbf{x}_S - \mathbf{x}_{S'})}$. The Hermiticity condition on the real-space hopping amplitudes guarantees $\mathbf{h}_{\mathbf{k}\sigma} = \mathbf{h}_{\mathbf{k}\sigma}^\dagger$ for every $\mathbf{k}$. The momentum-space Hamiltonian then takes the form

$$H = \sum_{\mathbf{k}\sigma} \mathbf{c}_{\mathbf{k}\sigma}^\dagger (\mathbf{h}_{\mathbf{k}\sigma} - \mu \mathbf{1}) \mathbf{c}_{\mathbf{k}\sigma} - \frac{U}{N_c} \sum_{S\mathbf{Q}} b_{S\mathbf{Q}}^\dagger b_{S\mathbf{Q}}, \quad (2)$$

where $\mathbf{c}_{\mathbf{k}\sigma} = (c_{\mathbf{k}1\sigma}, \dots, c_{\mathbf{k}N_S\sigma})^T$, with T denoting transposition, $\mathbf{1}$ is the identity matrix of the appropriate di-

mension (here $N_S \times N_S$), and $b_{S\mathbf{Q}}^\dagger = \sum_{\mathbf{k}} c_{\mathbf{k}+\mathbf{Q},S\uparrow}^\dagger c_{-\mathbf{k},S\downarrow}^\dagger$ creates an on-site singlet pair on orbital S with center-of-mass momentum $\mathbf{Q}$. Note that the orbital phases $e^{\pm i\mathbf{k} \cdot \mathbf{x}_S}$ carried by the Fourier convention cancel in the interaction by momentum conservation, so that the embedding $\{\mathbf{x}_S\}$ enters only through $\mathbf{h}_{\mathbf{k}\sigma}$; Appendix B discusses how the momentum-space representation transforms under a change of embedding representing the same real-space Hamiltonian. Since the interaction is diagonal in S , the pairing channels are labeled by the orbital index and the pairing field has N_S components. At the mean-field level, we decouple the interaction in the pairing channel by introducing the pairing amplitudes Δ_S , which at self-consistency are given by

$$\Delta_S = -U \langle c_{lS\downarrow} c_{lS\uparrow} \rangle, \quad (3)$$

independent of the unit-cell index l by translational invariance. We collect them into the pairing vector $\Delta = (\Delta_1, \dots, \Delta_{N_S})^T$, with $|\Delta|^2 = \Delta^\dagger \Delta = \sum_S |\Delta_S|^2$, and treat Δ as a variational parameter until it is fixed by the saddle-point condition below; its self-consistent value is denoted Δ_0 , with components Δ_{0S} . Dropping the term quadratic in the fluctuations about these averages replaces the interaction in Eq. (2) by

$$\sum_{\mathbf{k}S} (\Delta_S c_{\mathbf{k}S\uparrow}^\dagger c_{-\mathbf{k}S\downarrow}^\dagger + \Delta_S^* c_{-\mathbf{k}S\downarrow} c_{\mathbf{k}S\uparrow}) + N_c \frac{|\Delta|^2}{U}, \quad (4)$$

where the last term is the usual counterterm.

To determine the superfluid weight, we impose a uniform phase twist $\mathbf{q}$ on the pairing field, $\Delta_S \rightarrow \Delta_S e^{2i\mathbf{q} \cdot \mathbf{r}_{lS}}$, where the factor of 2 reflects the charge-two nature of the pair field. A gauge transformation removes this spatial phase from the pairing field and transfers the twist to the fermionic kinetic term, so that a pair is formed by $(\mathbf{k} + \mathbf{q}, \uparrow)$ and $(\mathbf{q} - \mathbf{k}, \downarrow)$ and the problem is again uniform [1, 31]. In this gauge, we introduce the Nambu spinor $\mathbf{C}(\mathbf{k}, \mathbf{q}) = (\mathbf{c}_{\mathbf{k}+\mathbf{q},\uparrow}, \mathbf{c}_{\mathbf{q}-\mathbf{k},\downarrow}^\dagger)^T$. The mean-field Hamiltonian and the corresponding Bogoliubov-de Gennes (BdG) Hamiltonian then take the form

$$H_0(\mathbf{q}) = \sum_{\mathbf{k}} \mathbf{C}^\dagger(\mathbf{k}, \mathbf{q}) \mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{q}) \mathbf{C}(\mathbf{k}, \mathbf{q}) + \sum_{\mathbf{k}} \text{tr}(\mathbf{h}_{\mathbf{q}-\mathbf{k},\downarrow} - \mu \mathbf{1}) + N_c \frac{|\Delta|^2}{U},$$

$$\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{q}) = \begin{bmatrix} \mathbf{h}_{\mathbf{k}+\mathbf{q},\uparrow} - \mu \mathbf{1} & \text{diag}(\Delta) \\ \text{diag}(\Delta)^\dagger & -(\mathbf{h}_{\mathbf{q}-\mathbf{k},\downarrow} - \mu \mathbf{1})^T \end{bmatrix}, \quad (5)$$

where, here and throughout, tr denotes the matrix trace over the orbital or Nambu-orbital space on which its argument acts (here the N_S -dimensional orbital space). The mean-field partition function is defined as $\mathcal{Z}_0(\mathbf{q}) = \text{Tr}_{\text{Fock}} e^{-\beta H_0(\mathbf{q})}$, with $\beta = 1/T$, where Tr_{Fock} runs over the full Fock space, and the thermodynamic potential per unit cell as $\Omega_0(\mathbf{q}, \Delta, \Delta^*) = -\frac{T}{N_c} \ln \mathcal{Z}_0(\mathbf{q})$. Diagonalizing $\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{q})$, with eigenvalues $E_p(\mathbf{k}, \mathbf{q})$, the thermo-

dynamic potential becomes

$$\Omega_0(\mathbf{q}, \Delta, \Delta^*) = \frac{|\Delta|^2}{U} + \frac{1}{N_c} \sum_{\mathbf{k}} \left\{ \text{tr}(\mathbf{h}_{\mathbf{q}-\mathbf{k}, \downarrow} - \mu \mathbf{1}) - T \sum_{p=1}^{2N_S} \ln \left[1 + e^{-E_p(\mathbf{k}, \mathbf{q})/T} \right] \right\}. \quad (6)$$

The constant trace term loses its apparent $\mathbf{q}$ dependence once $\mathbf{k}$ runs over the BZ, so that the twist enters Eq. (6) only through the quasiparticle spectrum.

At fixed $\mathbf{q}$, the self-consistent pairing vector is determined by the saddle-point equation

$$\frac{\partial \Omega_0(\mathbf{q}, \Delta, \Delta^*)}{\partial \Delta^*} = \mathbf{0}, \quad (7)$$

i.e. $\partial \Omega_0 / \partial \Delta_S^* = 0$ for every S , where Δ and Δ^* are treated as independent variables. By the Hellmann-Feynman theorem applied to Eq. (4), $\partial \Omega_0 / \partial \Delta_S^* = \Delta_S / U + \langle c_{IS\downarrow} c_{IS\uparrow} \rangle$, so that Eq. (7) is precisely the self-consistency condition (3); its complex conjugate, $\partial \Omega_0 / \partial \Delta = \mathbf{0}$, carries no further information. The imposed twist fixes only the winding $e^{2i\mathbf{q} \cdot \mathbf{r}_{IS}}$ of the pairing field; the underlying amplitudes are not held fixed, and Eq. (7) has to be solved at each $\mathbf{q}$. Its solution $\Delta_0(\mathbf{q})$ generally differs from $\Delta_0(\mathbf{0})$ not merely by an overall phase but in the ratios and the relative phases of the N_S components [23]. Substituting this solution back into the thermodynamic potential defines the potential along the self-consistent manifold, $\Omega(\mathbf{q}) = \Omega_0(\mathbf{q}, \Delta_0(\mathbf{q}), \Delta_0^*(\mathbf{q}))$.

The superfluid-weight tensor per unit volume is obtained from the curvature of the thermodynamic potential along the self-consistent solution,

$$V[D_s]_{ij} = N_c \frac{d^2 \Omega(\mathbf{q})}{dq_i dq_j} \Big|_{\mathbf{q}=\mathbf{0}}, \quad (8)$$

where V is the system volume. Because $\Delta_0(\mathbf{q})$ depends on the twist, evaluating the total derivative in Eq. (8) requires the expansion of Ω_0 to second order in the pairing field. Because the superconductor breaks the global $U(1)$ symmetry, a variation $\delta \Delta$ couples to $\delta \Delta$ as well as to $\delta \Delta^*$, and it is convenient to work in doubled complex coordinates. For a general pairing vector we write $\Psi = (\Delta, \Delta^*)^T$ and $\Omega_0(\mathbf{q}, \Psi) \equiv \Omega_0(\mathbf{q}, \Delta, \Delta^*)$; the self-consistent solution is $\Psi_0(\mathbf{q}) = (\Delta_0(\mathbf{q}), \Delta_0^*(\mathbf{q}))^T$, a variation about it is $\delta \Psi = (\delta \Delta, \delta \Delta^*)^T$, and throughout we abbreviate the untwisted saddle as $\Delta_0 \equiv \Delta_0(\mathbf{0})$ and $\Psi_0 \equiv \Psi_0(\mathbf{0})$. Expanding at fixed twist about the stationary point gives

$$\Omega_0(\mathbf{q}, \Psi_0(\mathbf{q}) + \delta \Psi) = \Omega(\mathbf{q}) + \mathbf{g}^\dagger(\mathbf{q}) \delta \Psi + \frac{1}{2} \delta \Psi^\dagger \mathcal{H}(\mathbf{q}) \delta \Psi, \quad (9)$$

up to second order in $\delta \Psi$, where $\mathbf{g}(\mathbf{q}, \Psi) = (\partial \Omega_0 / \partial \Delta^*, \partial \Omega_0 / \partial \Delta)^T$ is the gradient of Ω_0 at an arbitrary pairing vector Ψ and $\mathbf{g}(\mathbf{q}) \equiv \mathbf{g}(\mathbf{q}, \Psi_0(\mathbf{q}))$ is its on-shell value, which collects Eq. (7) and its conjugate

and therefore vanishes identically in $\mathbf{q}$, and the Hessian is the $2N_S \times 2N_S$ matrix of second derivatives with the same block ordering as Ψ ,

$$\mathcal{H}(\mathbf{q}) = \begin{pmatrix} \frac{\partial^2 \Omega_0}{\partial \Delta^* \partial \Delta^*} & \frac{\partial^2 \Omega_0}{\partial \Delta^* \partial \Delta} \\ \frac{\partial^2 \Omega_0}{\partial \Delta \partial \Delta^*} & \frac{\partial^2 \Omega_0}{\partial \Delta \partial \Delta} \end{pmatrix} \equiv \begin{pmatrix} \mathcal{A} & \mathcal{C} \\ \mathcal{C}^* & \mathcal{A}^* \end{pmatrix}, \quad (10)$$

all evaluated at $\Psi_0(\mathbf{q})$. The Hessian depends on $\mathbf{q}$ both through the twisted Hamiltonian and through $\Delta_0(\mathbf{q})$. Because Ω_0 is real, its four blocks are not independent: $\mathcal{A} = \mathcal{A}^\dagger$ and $\mathcal{C} = \mathcal{C}^T$, so that $\mathcal{H}(\mathbf{q})$ is Hermitian and its quadratic form on variations of the form $(\delta \Delta, \delta \Delta^*)^T$ is the real second variation of Ω_0 . At a stable saddle it is positive semidefinite, $\mathcal{H}(\mathbf{q}) \succeq 0$, and its spectral decomposition may be used freely.

Differentiating the gap equation with respect to the phase twist relates the response of the pairing vector to this Hessian. Denoting the response by $\mathbf{F}_i = \partial_{q_i} \Psi_0(\mathbf{q})|_{\mathbf{q}=\mathbf{0}} = (\partial_{q_i} \Delta_0(\mathbf{q}), \partial_{q_i} \Delta_0^*(\mathbf{q}))^T|_{\mathbf{q}=\mathbf{0}}$ and the mixed derivative by $\mathbf{B}_i = \partial_{q_i} \mathbf{g}(\mathbf{q}, \Psi)|_{\mathbf{q}=\mathbf{0}, \Psi=\Psi_0}$, i.e. the explicit $\mathbf{q}$ derivative of the gradient at fixed pairing vector, which is not the derivative of $\mathbf{g}(\mathbf{q})$ since the latter vanishes identically, the total derivative of $\mathbf{g}(\mathbf{q}, \Psi_0(\mathbf{q})) = \mathbf{0}$ gives

$$\mathcal{H}(\mathbf{0}) \mathbf{F}_i = -\mathbf{B}_i. \quad (11)$$

Global $U(1)$ invariance, $\Omega_0(\mathbf{q}, e^{i\phi} \Delta, e^{-i\phi} \Delta^*) = \Omega_0(\mathbf{q}, \Delta, \Delta^*)$, fixes the pairing vector only up to a common phase. For an infinitesimal global phase rotation of the untwisted saddle, the corresponding variation is $\delta \Psi = \phi \mathbf{u}_0$, where $\mathbf{u}_0 = (i\Delta_0, -i\Delta_0^*)^T$. Inserting $\delta \Psi$ into Eq. (9) and using $\mathbf{g} = \mathbf{0}$ gives

$$\mathcal{H}(\mathbf{0}) \mathbf{u}_0 = \mathbf{0}, \quad \langle \mathbf{u}_0 | \mathbf{u}_0 \rangle = 2|\Delta_0|^2, \quad (12)$$

i.e. the global phase rotation is a zero mode of the Hessian [24]. This is the thermodynamic form of the Goldstone theorem, and $\mathbf{u}_0$ reappears in Sec. III B as the Goldstone direction of the pair-fluctuation kernel. The component of $\mathbf{F}_i$ along $\mathbf{u}_0$ is left undetermined by Eq. (11), reflecting the freedom to choose the global phase, but it drops out of every physical quantity below.

The first total derivative of $\Omega(\mathbf{q})$ equals the partial derivative at frozen pairing field, because the implicit contributions vanish by the gap equation. This is not the case for the second derivative, since one then differentiates the gradient of Ω_0 with respect to $\mathbf{q}$, which the gap equation does not eliminate. Using the chain rule together with Eq. (11), we obtain

$$V[D_s]_{ij} = N_c \frac{\partial^2 \Omega_0}{\partial q_i \partial q_j} \Big|_{\mathbf{q}=\mathbf{0}, \Psi=\Psi_0} - N_c \mathbf{F}_i^\dagger \mathcal{H}(\mathbf{0}) \mathbf{F}_j. \quad (13)$$

The first term is the curvature evaluated at frozen pairing amplitudes, $V[D_{\text{frozen}}]_{ij} \equiv N_c \frac{\partial^2 \Omega_0}{\partial q_i \partial q_j} \Big|_{\mathbf{q}=\mathbf{0}, \Psi=\Psi_0}$, which contains the usual band-dispersion and Bloch-geometric

contributions. The second term accounts for the self-consistent relaxation of the pairing field under the phase twist: as the twist is switched on, the pairing vector rearranges as $\Psi_0(\mathbf{q}) = \Psi_0 + \sum_i q_i \mathbf{F}_i + O(\mathbf{q}^2)$, and the system lowers its energy by doing so. Since $\mathcal{H}(\mathbf{0}) \succeq 0$ at a stable saddle, the quadratic form $\mathbf{F}_i^\dagger \mathcal{H}(\mathbf{0}) \mathbf{F}_j$ is positive semidefinite, and it enters Eq. (13) with a negative sign. The relaxation therefore lowers the superfluid weight, $\mathbf{D}_s \preceq \mathbf{D}_{\text{frozen}}$ [24]. Equation (13) also has a variational reading: applying the same chain rule along an arbitrary smooth path $\tilde{\Psi}(\mathbf{q})$ through the saddle, with tangent $\tilde{\mathbf{F}}_i$, gives $N_c d^2 \Omega_0(\mathbf{q}, \tilde{\Psi}(\mathbf{q}))/dq_i dq_j|_0 = V[D_s]_{ij} + N_c (\tilde{\mathbf{F}}_i - \mathbf{F}_i)^\dagger \mathcal{H}(\mathbf{0}) (\tilde{\mathbf{F}}_j - \mathbf{F}_j)$, so that every trial path, the frozen one included, overestimates the superfluid weight.

A corresponding relaxation correction was obtained in Ref. [24] at fixed particle number using a gauge-fixed parametrization of the order parameter. The two treatments differ in the ensemble and in the gauge, and neither difference affects the content of Eq. (13). At fixed particle number one differentiates the free energy $F = \Omega + \mu N$, so that $\mu = \mu(\mathbf{q})$ becomes an additional self-consistent variable determined by $N = -\partial \Omega / \partial \mu$ [31]. Treating it in the same way as the pairing amplitudes above, with $\Omega(\mathbf{q}, \mu)$ the gap-relaxed potential of Eq. (13) at fixed μ , gives $\frac{d^2 F}{dq_i dq_j}|_N = \frac{d^2 \Omega}{dq_i dq_j}|_\mu - \frac{(\partial_{q_i} \partial_\mu \Omega)(\partial_{q_j} \partial_\mu \Omega)}{\partial_\mu^2 \Omega}$. The second term is the relaxation of the chemical potential. It carries the opposite sign to the gap relaxation, because $\partial_\mu^2 \Omega = -\partial N / \partial \mu < 0$, so that the fixed- N stiffness is never smaller than the fixed- μ one. The mixed derivative $\partial_{q_i} \partial_\mu \Omega$ vanishes at $\mathbf{q} = \mathbf{0}$ whenever $\Omega(\mathbf{q}) = \Omega(-\mathbf{q})$, for instance in the presence of time-reversal or inversion symmetry, in which case the two ensembles give the same superfluid weight [24]. As for the gauge, Ref. [24] fixes the global phase by choosing one pairing amplitude to be real and positive, which removes the associated phase degree of freedom and renders the reduced Hessian invertible. Here we retain the full doubled complex pairing space, including both amplitude and phase variations. Consequently $\mathcal{H}(\mathbf{0})$ contains the zero mode $\mathbf{u}_0$ of Eq. (12) and is not invertible in the full pairing space, which Eq. (13) never requires.

III. EFFECTIVE-ACTION FORMULATION

We now turn to the effective-action description of pairing fluctuations. Here no twist is imposed: spatial variations enter through the momentum dependence of the fluctuations about the untwisted saddle, and a fermionic twist $\mathbf{q}$ corresponds, by a gauge transformation, to a pairing field of center-of-mass momentum $\mathbf{Q} = 2\mathbf{q}$ (Appendix C). This is what allows both terms of Eq. (13) to be read off from the momentum dependence of the Gaussian fluctuation kernel.

A. Effective action for the pairing field

We start from the coherent-state path-integral representation of the partition function, $\mathcal{Z} = \int \mathcal{D}[\bar{\psi}, \psi] e^{-\mathcal{S}[\bar{\psi}, \psi]}$, where $\psi_{k\sigma} = (\psi_{k1\sigma}, \dots, \psi_{kN_S\sigma})^T$ are the Grassmann counterparts of the operators $\mathbf{c}_{k\sigma}$ of Eq. (2), $\bar{\psi}_{k\sigma}$ is the independent conjugate field, and $k = (\mathbf{k}, i\omega_n)$ with $\omega_n = (2n + 1)\pi T$. In these variables the untwisted action corresponding to Eq. (2) is $\mathcal{S} = \sum_{k\sigma} \bar{\psi}_{k\sigma} (-i\omega_n \mathbf{1} + \mathbf{h}_{k\sigma} - \mu \mathbf{1}) \psi_{k\sigma} - \frac{U}{\beta N_c} \sum_{QS} \bar{b}_S(Q) b_S(Q)$, with $\bar{b}_S(Q) = \sum_k \bar{\psi}_{k+Q, S\uparrow} \bar{\psi}_{-k, S\downarrow}$ the Grassmann counterpart of $b_S^\dagger(\mathbf{Q})$, and $Q = (\mathbf{Q}, i\nu_m)$ with $\nu_m = 2\pi mT$. The quartic term is decoupled by a Hubbard-Stratonovich transformation in the pairing channel, channel by channel and momentum by momentum. Introducing the pairing fields $\Phi_S(Q)$ through $\Phi_{lS}(\tau) = \frac{1}{\sqrt{\beta N_c}} \sum_Q e^{i(\mathbf{Q} \cdot \mathbf{r}_{lS} - \nu_m \tau)} \Phi_S(Q)$, which couple linearly to $\bar{b}_S(Q)$, gives

$$\mathcal{S}_{\text{HS}} = \sum_{k\sigma} \bar{\psi}_{k\sigma} (-i\omega_n \mathbf{1} + \mathbf{h}_{k\sigma} - \mu \mathbf{1}) \psi_{k\sigma} + \sum_{QS} \frac{|\Phi_S(Q)|^2}{U} + \frac{1}{\sqrt{\beta N_c}} \sum_{kQS} [\Phi_S(Q) \bar{\psi}_{k+Q, S\uparrow} \bar{\psi}_{-k, S\downarrow} + \text{c.c.}], \quad (14)$$

with $\mathcal{Z} = \int \mathcal{D}[\bar{\psi}, \psi, \Phi, \Phi^*] e^{-\mathcal{S}_{\text{HS}}}$, where $\Phi \equiv \{\Phi_S(Q)\}$ denotes the collection of pairing fields. Gaussian integration over Φ returns the quartic action above, which fixes the relative sign of the last two terms of Eq. (14); with this choice a uniform static field $\Phi_S(Q) = \sqrt{\beta N_c} \Delta_S \delta_{Q0}$ reproduces Eq. (4), where δ_{Q0} is a Kronecker delta. For any static field, $\Phi_S(Q) \propto \delta_{\nu_m, 0}$, the τ dependence of $\Phi_{lS}(\tau)$ drops out, and we then write simply Φ_{lS} ; in particular the uniform static field above gives $\Phi_{lS} = \Delta_S$, and the $\sqrt{\beta N_c}$ in the Fourier convention is what makes the real-space field carry the normalization of Eq. (3).

Equation (14) is quadratic in the fermions, which can therefore be integrated out exactly. Collecting them into the Nambu spinor $\mathbf{C}_k = (\psi_{k\uparrow}, \bar{\psi}_{-k\downarrow})^T$, the Grassmann counterpart of the spinor of Sec. II at zero twist, the fermionic part of Eq. (14) reads $-\sum_{kk'} \bar{\mathbf{C}}_k \mathbf{G}^{-1}[\Phi]_{kk'} \mathbf{C}_{k'}$, with

$$\mathbf{G}^{-1}[\Phi]_{kk'} = \delta_{kk'} [i\omega_n \mathbf{1} - \mathbf{h}_{\text{BdG}}^{(N)}(\mathbf{k})] - \sum_{\alpha=1}^{2N_S} \frac{\Phi_\alpha(k - k')}{\sqrt{\beta N_c}} \mathbf{V}_\alpha \quad (15)$$

the inverse Nambu propagator, where $\mathbf{h}_{\text{BdG}}^{(N)}$ is the normal-state BdG matrix, i.e. Eq. (5) at $\Delta = \mathbf{0}$ and $\mathbf{q} = \mathbf{0}$, the doubled pairing field is $\Phi(Q) = (\Phi_1(Q), \dots, \Phi_{N_S}(Q), \Phi_1^*(-Q), \dots, \Phi_{N_S}^*(-Q))^T$ with components $\Phi_\alpha(Q)$, $\alpha = 1, \dots, 2N_S$, and the pairing vertices $\mathbf{V}_\alpha$ are defined in Appendix A. A pairing field of four-momentum Q thus connects two fermionic propagators whose momenta and frequencies differ by Q . Performing the Grassmann integration gives the exact

bosonic action

$$\mathcal{S}_{\text{eff}}[\Phi] = \frac{1}{U} \sum_{QS} |\Phi_S(Q)|^2 - \text{Tr} \ln [-\mathbf{G}^{-1}[\Phi]], \quad (16)$$

with $\mathcal{Z} = \int \mathcal{D}[\Phi, \Phi^*] e^{-\mathcal{S}_{\text{eff}}[\Phi]}$ and Tr the functional trace over momentum, Matsubara frequency and Nambu-orbital indices; this is the only meaning of Tr in what follows.

B. Static and uniform Gaussian fluctuations

Below T_c , we expand about the uniform self-consistent saddle as

$$\Phi_S(Q) = \sqrt{\beta N_c} \Delta_{0S} \delta_{Q0} + \eta_S(Q), \quad (17)$$

and collect the fluctuations $\boldsymbol{\eta}(Q) = (\eta_1(Q), \dots, \eta_{N_s}(Q))^T$ into the doubled field $\delta\Phi(Q) = (\boldsymbol{\eta}(Q), \boldsymbol{\eta}^*(-Q))^T$, so that the doubled field of Sec. III A is $\Phi(Q) = \sqrt{\beta N_c} \Psi_0 \delta_{Q0} + \delta\Phi(Q)$. The two sectors of $\delta\Phi(Q)$ are not independent but related by

$$\delta\Phi(-Q) = \boldsymbol{\tau}_1 \delta\Phi^*(Q), \quad \boldsymbol{\tau}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (18)$$

where $\boldsymbol{\tau}_1$ exchanges the two sectors of the doubled pairing space. Expanding the effective action about the saddle gives $\mathcal{S}_{\text{eff}} = \mathcal{S}_{\text{eff}}^{(0)} + \mathcal{S}_{\text{eff}}^{(2)} + \dots$, where $\mathcal{S}_{\text{eff}}^{(0)} = \beta N_c \Omega_0(\mathbf{0}, \Psi_0)$ is the mean-field value and the linear term vanishes by the saddle-point condition. Since the saddle is translationally invariant, the quadratic action is diagonal in four-momentum,

$$\mathcal{S}_{\text{eff}}^{(2)} = \frac{1}{2} \sum_Q \delta\Phi^\dagger(Q) \mathbf{M}(Q) \delta\Phi(Q), \quad (19)$$

which defines the Gaussian pairing-fluctuation kernel $\mathbf{M}(Q)$, equivalently the inverse fluctuation propagator.

Equation (19) contains the thermodynamic Hessian of Sec. II as its zero-four-momentum limit. For a static and spatially uniform fluctuation, $Q = (\mathbf{0}, 0)$, the Fourier convention gives $\delta\Phi(\mathbf{0}, 0) = \sqrt{\beta N_c} \delta\Psi$ with $\delta\Psi$ the uniform variation of Ψ_0 , so that only the single term $Q = (\mathbf{0}, 0)$ survives in Eq. (19) and $\mathcal{S}_{\text{eff}}^{(2)} = \frac{\beta N_c}{2} \delta\Psi^\dagger \mathbf{M}(\mathbf{0}, 0) \delta\Psi$. On the other hand, for static and spatially uniform configurations the effective action is the mean-field potential itself, $\mathcal{S}_{\text{eff}} = \beta N_c \Omega_0(\mathbf{0}, \Psi)$, whose expansion (9) gives $\mathcal{S}_{\text{eff}}^{(2)} = \frac{\beta N_c}{2} \delta\Psi^\dagger \mathcal{H}(\mathbf{0}) \delta\Psi$. Comparing the two quadratic forms yields

$$\mathbf{M}(\mathbf{0}, 0) = \mathcal{H}(\mathbf{0}). \quad (20)$$

The thermodynamic Hessian is therefore not merely analogous to the fluctuation kernel: it is that kernel at zero momentum and zero frequency. At finite momentum or frequency $\mathbf{M}(Q)$ carries spatial and dynamical information that is absent from $\mathcal{H}(\mathbf{0})$, so the identification is

static. A microscopic derivation of Eq. (20) from the effective action is given in Appendix A.

We now restrict to static pairing fluctuations and define $\mathbf{M}(\mathbf{Q}) \equiv \mathbf{M}(\mathbf{Q}, i\nu_m = 0)$. Starting from the microscopic representation of $\mathbf{M}(Q)$ derived in Appendix A, performing the Matsubara sum and evaluating the trace in the eigenbasis of $\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{0})$ gives the static kernel in a form that can be evaluated directly,

$$[M(\mathbf{Q})]_{\alpha\alpha'} = \frac{\delta_{\alpha\alpha'}}{U} + \frac{1}{N_c} \sum_{\mathbf{k}} \sum_{pp'} \langle p', \mathbf{k} | \mathbf{V}_\alpha^\dagger | p, \mathbf{k} + \mathbf{Q} \rangle \times \langle p, \mathbf{k} + \mathbf{Q} | \mathbf{V}_{\alpha'} | p', \mathbf{k} \rangle \frac{f[E_p(\mathbf{k} + \mathbf{Q})] - f[E_{p'}(\mathbf{k})]}{E_p(\mathbf{k} + \mathbf{Q}) - E_{p'}(\mathbf{k})}, \quad (21)$$

where $|p, \mathbf{k}\rangle$ and $E_p(\mathbf{k}) \equiv E_p(\mathbf{k}, \mathbf{0})$ are the eigenvectors and eigenvalues of the untwisted BdG matrix $\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{0})$, f is the Fermi function, and the ratio is replaced by its derivative $f'(E_p)$ for coincident energies. Since $\mathbf{h}_{\mathbf{k}+\mathbf{Q}, \sigma}$ is known analytically, Eq. (21) can be evaluated at any real $\mathbf{Q}$, which need not be commensurate with the $\mathbf{k}$ grid.

The static kernel is Hermitian for real $\mathbf{Q}$, and the constraint (18) on the doubled field implies the particle-hole relation

$$\boldsymbol{\tau}_1 \mathbf{M}^*(\mathbf{Q}) \boldsymbol{\tau}_1 = \mathbf{M}(-\mathbf{Q}). \quad (22)$$

Differentiating Eq. (22) at $\mathbf{Q} = \mathbf{0}$ gives $\boldsymbol{\tau}_1 (\partial_i \mathbf{M})^* \boldsymbol{\tau}_1 = -\partial_i \mathbf{M}$ and $\boldsymbol{\tau}_1 (\partial_i \partial_j \mathbf{M})^* \boldsymbol{\tau}_1 = +\partial_i \partial_j \mathbf{M}$, where $\partial_i \equiv \partial/\partial Q_i|_{\mathbf{Q}=\mathbf{0}}$ denotes the pair-momentum derivative at zero pair momentum. Throughout, a bare ∂_i carries this meaning; derivatives at general $\mathbf{Q}$ are written ∂_{Q_i} , derivatives with respect to the fermionic twist ∂_{q_i} , and real-space gradients ∇_i . By Eq. (20), the zero mode $\mathbf{u}_0$ of Eq. (12) is also a zero mode of the static kernel, $\mathbf{M}(\mathbf{0}) \mathbf{u}_0 = \mathbf{0}$, at every temperature at which the symmetry is broken. We normalize it as $\mathbf{w}_0 = \mathbf{u}_0/(\sqrt{2}|\Delta_0|)$ and complete the eigenbasis with orthonormal massive modes satisfying

$$\mathbf{M}(\mathbf{0}) \mathbf{w}_n = \lambda_n \mathbf{w}_n, \quad n > 0, \quad (23)$$

with $\langle \mathbf{w}_m | \mathbf{w}_n \rangle = \delta_{mn}$. The λ_n are the static stiffnesses of the amplitude and relative-phase channels; the latter correspond to Leggett modes [32, 33] when their dynamical continuation is well defined, although no statement about collective-mode dynamics is implied here. At a stable saddle $\lambda_n > 0$, and in the following we assume that the $U(1)$ Goldstone mode is the only zero mode of $\mathbf{M}(\mathbf{0})$; at a secondary instability one of the massive modes may itself soften.

C. Goldstone projection and low- $\mathbf{Q}$ expansion

The eigenvectors $\{\mathbf{w}_0, \mathbf{w}_n\}$ with $n > 0$ form a complete basis in the doubled pairing space. We use this fixed zero-momentum basis to describe a slowly varying static

fluctuation, writing $\delta\Phi(\mathbf{Q}) \equiv \delta\Phi(\mathbf{Q}, i\nu_m = 0)$ and decomposing

$$\delta\Phi(\mathbf{Q}) = \theta(\mathbf{Q})\mathbf{w}_0 + \sum_{n>0} \chi_n(\mathbf{Q})\mathbf{w}_n, \quad (24)$$

where $\theta(\mathbf{Q})$ is the Goldstone-mode amplitude and $\chi(\mathbf{Q})$ collects the massive ones. In this basis the static kernel becomes a block matrix with elements $M_{00}(\mathbf{Q}) = \langle \mathbf{w}_0 | \mathbf{M}(\mathbf{Q}) | \mathbf{w}_0 \rangle$, $[M_{\perp 0}(\mathbf{Q})]_n = \langle \mathbf{w}_n | \mathbf{M}(\mathbf{Q}) | \mathbf{w}_0 \rangle$, $[M_{0\perp}(\mathbf{Q})]_n = \langle \mathbf{w}_0 | \mathbf{M}(\mathbf{Q}) | \mathbf{w}_n \rangle$, and $[M_{\perp\perp}(\mathbf{Q})]_{mn} = \langle \mathbf{w}_m | \mathbf{M}(\mathbf{Q}) | \mathbf{w}_n \rangle$. Collecting the amplitudes into $\zeta(\mathbf{Q}) = (\theta(\mathbf{Q}), \chi(\mathbf{Q}))^T$, the quadratic action reads

$$\mathcal{S}_{\text{eff}}^{(2)} = \frac{1}{2} \sum_{\mathbf{Q}} \zeta^\dagger(\mathbf{Q}) \begin{bmatrix} M_{00}(\mathbf{Q}) & \mathbf{M}_{0\perp}(\mathbf{Q}) \\ \mathbf{M}_{\perp 0}(\mathbf{Q}) & \mathbf{M}_{\perp\perp}(\mathbf{Q}) \end{bmatrix} \zeta(\mathbf{Q}). \quad (25)$$

The Goldstone mode is coupled to the massive modes through the off-diagonal blocks, which vanish at $\mathbf{Q} = \mathbf{0}$ because $\{\mathbf{w}_0, \mathbf{w}_n\}$ diagonalize $\mathbf{M}(\mathbf{0})$. Since $\mathbf{M}_{\perp\perp}(\mathbf{0})$ has eigenvalues $\lambda_n > 0$, that block remains invertible for sufficiently small $\mathbf{Q}$, and the massive amplitudes can be integrated out. At quadratic order this is equivalent to eliminating them through their stationary values, $\chi(\mathbf{Q}) = -\mathbf{M}_{\perp\perp}^{-1}(\mathbf{Q})\mathbf{M}_{\perp 0}(\mathbf{Q})\theta(\mathbf{Q})$, which leaves $\mathcal{S}_{\text{eff}}^{(2)} = \frac{1}{2} \sum_{\mathbf{Q}} \theta^*(\mathbf{Q}) M_{\text{eff}}(\mathbf{Q}) \theta(\mathbf{Q})$ with

$$M_{\text{eff}}(\mathbf{Q}) = M_{00}(\mathbf{Q}) - \mathbf{M}_{0\perp}(\mathbf{Q})\mathbf{M}_{\perp\perp}^{-1}(\mathbf{Q})\mathbf{M}_{\perp 0}(\mathbf{Q}). \quad (26)$$

Thus $M_{\text{eff}}(\mathbf{Q})$ is the Schur complement of the massive sector, and represents the energy cost of a slowly varying Goldstone-mode fluctuation after all massive pairing degrees of freedom have relaxed. Since $\mathbf{M}_{\perp\perp}(\mathbf{Q})$ is positive definite in a stable phase, so is its inverse, and the subtracted term in Eq. (26) is non-negative: the relaxation can only lower the Goldstone stiffness relative to its frozen-pairing value, in the same way that two springs in series are softer than either spring alone.

We now expand $M_{\text{eff}}(\mathbf{Q})$ about $\mathbf{Q} = \mathbf{0}$. The zeroth-order term vanishes because $\mathbf{w}_0$ is the Goldstone mode. Since $\mathbf{w}_0$ is invariant under τ_1 complex conjugation, Eq. (22) implies that $M_{00}(\mathbf{Q})$ is real and even in $\mathbf{Q}$, so that $\langle \mathbf{w}_0 | \partial_i \mathbf{M} | \mathbf{w}_0 \rangle = 0$ without invoking inversion or time-reversal symmetry. Each mixed block is of order $\mathbf{Q}$, so that at order $\mathbf{Q}^2$ only $[M_{\perp\perp}^{-1}(\mathbf{0})]_{mn} = \delta_{mn}/\lambda_n$ is needed and derivatives of $\mathbf{M}_{\perp\perp}^{-1}$ do not enter. Equation (26) then gives $M_{\text{eff}}(\mathbf{Q}) = \sum_{ij} Q_i Q_j \mathcal{Q}_{ij} + O(\mathbf{Q}^4)$, with

$$\mathcal{Q}_{ij} = \frac{1}{2} \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle - \sum_{n>0} \frac{\text{Re} [\langle \mathbf{w}_0 | \partial_i \mathbf{M} | \mathbf{w}_n \rangle \langle \mathbf{w}_n | \partial_j \mathbf{M} | \mathbf{w}_0 \rangle]}{\lambda_n}. \quad (27)$$

Here Re denotes the real part, and $\mathcal{Q}_{ij}$ is real and symmetric. The first term is the direct curvature of the Goldstone sector, and the second accounts for its coupling to, and relaxation through, the massive pairing modes. If

the uniform saddle is locally stable against static fluctuations at small finite momentum, then $\mathbf{M}(\mathbf{Q}) \succeq 0$ near $\mathbf{Q} = \mathbf{0}$ and consequently $\mathcal{Q} \succeq 0$; a negative eigenvalue of $\mathcal{Q}$ signals an instability toward finite pair momentum.

Equation (27) is nothing but nondegenerate Rayleigh-Schrödinger perturbation theory for the branch $\lambda_0(\mathbf{Q})$ of $\mathbf{M}(\mathbf{Q})$ that evolves continuously from the Goldstone mode, with $\lambda_0(\mathbf{0}) = 0$: treating $\mathbf{M}(\mathbf{0})$ as the unperturbed operator and $\mathbf{M}(\mathbf{Q}) - \mathbf{M}(\mathbf{0})$, expanded to second order in $\mathbf{Q}$, as the perturbation, the first- and second-order shifts of that level, with energy denominators $\lambda_0(\mathbf{0}) - \lambda_n = -\lambda_n$, reproduce the two terms of Eq. (27) once symmetrized in i and j . Since $\lambda_0(\mathbf{Q})$ is even in $\mathbf{Q}$ by Eq. (22), the omitted terms are of order $\mathbf{Q}^4$, and therefore

$$\mathcal{Q}_{ij} = \frac{1}{2} \frac{\partial^2 \lambda_0(\mathbf{Q})}{\partial Q_i \partial Q_j} \Big|_{\mathbf{Q}=\mathbf{0}}, \quad (28)$$

which is a basis-independent definition of $\mathcal{Q}_{ij}$: an eigenvalue is invariant under any unitary change of the channel basis, including a $\mathbf{Q}$ -dependent one, whereas the two terms of Eq. (27) separately are not. The Schur complement along the fixed direction $\mathbf{w}_0$ and the exact eigenvalue differ only at order $\mathbf{Q}^4$.

D. Quantum geometry of Goldstone modes

We next identify the geometric origin of the relaxation term. The eigenvector $\mathbf{w}_0(\mathbf{Q})$ of the soft branch satisfies $\mathbf{M}(\mathbf{Q})\mathbf{w}_0(\mathbf{Q}) = \lambda_0(\mathbf{Q})\mathbf{w}_0(\mathbf{Q})$, with $\mathbf{w}_0(\mathbf{0}) = \mathbf{w}_0$. Differentiating at $\mathbf{Q} = \mathbf{0}$, using $\lambda_0(\mathbf{0}) = 0$ and $\partial_i \lambda_0 = \langle \mathbf{w}_0 | \partial_i \mathbf{M} | \mathbf{w}_0 \rangle = 0$, and projecting onto a massive eigenmode gives

$$\langle \mathbf{w}_n | \partial_i \mathbf{M} | \mathbf{w}_0 \rangle = -\lambda_n \langle \mathbf{w}_n | \partial_i \mathbf{w}_0 \rangle, \quad n > 0. \quad (29)$$

Inserting this into Eq. (27) converts the second-order eigenvalue shift into a stiffness-weighted overlap of eigenvector derivatives. As the pair momentum is varied, the soft eigenvector $\mathbf{w}_0(\mathbf{Q})$ traces a curve in the doubled pairing space. Resolving its tangent along the massive branches $\mathbf{w}_n(\mathbf{Q})$, i.e. the eigenvectors of $\mathbf{M}(\mathbf{Q})$ that evolve from $\mathbf{w}_n$, defines the mode-resolved pair quantum metric of the Goldstone mode,

$$g_{0\mathbf{Q}}^{nij} = 2 \text{Re} [\langle \partial_{Q_i} \mathbf{w}_0(\mathbf{Q}) | \mathbf{w}_n(\mathbf{Q}) \rangle \langle \mathbf{w}_n(\mathbf{Q}) | \partial_{Q_j} \mathbf{w}_0(\mathbf{Q}) \rangle]. \quad (30)$$

Equation (30) is a metric on the manifold of pairing configurations swept out by $\mathbf{Q}$ rather than on the Bloch manifold swept out by $\mathbf{k}$. Only its value at zero pair momentum enters below, and with it Eq. (27) becomes

$$\mathcal{Q}_{ij} = \frac{1}{2} \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle - \frac{1}{2} \sum_{n>0} \lambda_n g_{00}^{nij}. \quad (31)$$

The relaxation correction is therefore the stiffness-weighted pair quantum metric $\sum_{n>0} \lambda_n g_{00}^{nij}$. It is gauge

invariant under separate rephasings of $\mathbf{w}_0$ and $\mathbf{w}_n$, because $\langle \mathbf{w}_0 | \mathbf{w}_n \rangle = 0$, and it requires the Goldstone branch to remain nondegenerate near $\mathbf{Q} = \mathbf{0}$, so that $\mathbf{w}_0(\mathbf{Q})$ can be continued smoothly. Three geometric objects should be distinguished here. The Bloch quantum metric of the band structure enters only indirectly, through the inter-band matrix elements of $\partial_i \partial_j \mathbf{M}$ [24]. The unweighted sum $\sum_{n>0} g_{00}^{nij} = 2 \text{Re} [\langle \partial_i \mathbf{w}_0 | (1 - |\mathbf{w}_0\rangle\langle \mathbf{w}_0|) | \partial_j \mathbf{w}_0 \rangle]$ is the full Fubini-Study metric of the soft eigenvector, a property of that eigenvector alone. What enters the superfluid weight is neither of these, but the sum in which each massive channel is weighted by its own static stiffness λ_n [34].

IV. SUPERFLUID WEIGHT

Two facts connect the thermodynamic formulation of Sec. II to the fluctuation kernel of Sec. III. The first is the field-theoretical reading of the twist. The twisted mean-field problem of Sec. II is not a new theory: it is the untwisted exact action (16) evaluated on the particular configuration in which the pairing field is a single static Fourier mode of momentum $2\mathbf{q}$, i.e. in which it winds in real space,

$$\begin{aligned} \Phi_S(Q) &= \sqrt{\beta N_c} \Delta_S \delta_{\mathbf{Q}, 2\mathbf{q}} \delta_{\nu_m, 0}, \quad \Phi_{lS} = \Delta_S e^{2i\mathbf{q} \cdot \mathbf{r}_{lS}}, \\ \Phi(\mathbf{r}) &= \mathbf{U}_{\mathbf{q}}(\mathbf{r}) \Psi, \quad \mathbf{U}_{\mathbf{q}}(\mathbf{r}) = e^{2i\mathbf{q} \cdot \mathbf{r} \tau_3}, \quad \tau_3 = \text{diag}(\mathbf{1}, -\mathbf{1}), \end{aligned} \quad (32)$$

where $\Phi(\mathbf{r})$ denotes the doubled real-space pairing field at position $\mathbf{r} = \mathbf{r}_{lS}$, the conjugate component carrying the opposite phase, and $\Psi = (\Delta, \Delta^*)^T$ is an arbitrary uniform pairing vector. By definition of the saddle-point action,

$$\mathcal{S}_{\text{eff}}[\mathbf{U}_{\mathbf{q}} \Psi] = \beta N_c \Omega_0(\mathbf{q}, \Psi) \quad (33)$$

for every Ψ , which at $\mathbf{q} = \mathbf{0}$ and $\Psi = \Psi_0$ reduces to $\mathcal{S}_{\text{eff}}^{(0)}$ of Sec. III B, since $\mathbf{U}_0 = \mathbf{1}$. The gauge transformation of Sec. II, which moved the phase from the pairing field onto the fermions, is therefore the statement that the fermionic determinant of this configuration is that of $\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{q})$. Every $\mathbf{q}$ derivative of $\Omega_0(\mathbf{q}, \cdot)$ is consequently a functional derivative of the untwisted action along the winding configuration, and such derivatives at Ψ_0 are given by the kernel of Sec. III B. The second fact is the identification (20) of the thermodynamic Hessian with that kernel at zero momentum and zero frequency. With these two facts both terms of Eq. (13) can be evaluated.

Using Eq. (32) together with translational invariance, as detailed in Appendix C, the mixed derivative introduced in Sec. II becomes

$$\mathbf{B}_i = -2i(\partial_i \mathbf{M}) \mathbf{u}_0. \quad (34)$$

Projecting Eq. (11) onto a massive eigenmode $\mathbf{w}_n$ and using Eqs. (34) and (29) gives, for every $n > 0$,

$$\lambda_n \langle \mathbf{w}_n | \mathbf{F}_i \rangle = -2\sqrt{2}i |\Delta_0| \lambda_n \langle \mathbf{w}_n | \partial_i \mathbf{w}_0 \rangle. \quad (35)$$

At a stable saddle $\lambda_n > 0$, so the stiffness cancels on both sides and the response is fixed up to its undetermined Goldstone component,

$$\mathbf{P}_{\perp} \mathbf{F}_i = -2\sqrt{2}i |\Delta_0| \mathbf{P}_{\perp} | \partial_i \mathbf{w}_0 \rangle, \quad \mathbf{P}_{\perp} = \mathbf{1} - |\mathbf{w}_0\rangle\langle \mathbf{w}_0|. \quad (36)$$

That is, the self-consistent response of the pairing vector to the twist is the transverse part of the pair-momentum derivative of the soft eigenvector: as $\mathbf{Q} = 2\mathbf{q}$ is turned on, the gaps follow the rotation of $\mathbf{w}_0(\mathbf{Q})$. Both sides of Eq. (36) carry the same arbitrariness: $\mathbf{F}_i$ is undetermined along $\mathbf{u}_0$ because of global $U(1)$, and $\mathbf{w}_0(\mathbf{Q})$ is defined only up to a $\mathbf{Q}$ -dependent phase. The projector $\mathbf{P}_{\perp}$ removes both, so the identification is between gauge-covariant objects. Contracting Eq. (36) with $\mathcal{H}(\mathbf{0})$, which annihilates $\mathbf{u}_0$, gives the relaxation term in geometric form,

$$-\mathbf{F}_i^{\dagger} \mathcal{H}(\mathbf{0}) \mathbf{F}_j = -4 |\Delta_0|^2 \sum_{n>0} \lambda_n g_{00}^{nij}, \quad (37)$$

the stiffness-weighted pair quantum metric of Eq. (31). This provides a geometric interpretation of the relaxation correction of Ref. [24].

The same gauge relation connects the frozen-pairing curvature to the direct Goldstone-sector curvature,

$$V[D_{\text{frozen}}]_{ij} = 4N_c |\Delta_0|^2 \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle, \quad (38)$$

as shown in Appendix C. Combining Eqs. (38) and (37) with Eq. (13), and using Eq. (31), gives

$$V[D_s]_{ij} = 8N_c |\Delta_0|^2 \mathcal{Q}_{ij}. \quad (39)$$

The prefactor 8 is the product of the Nambu norm $\langle \mathbf{u}_0 | \mathbf{u}_0 \rangle / |\Delta_0|^2 = 2$ of the Goldstone vector and a factor 4 from $\nabla_i \phi = 2q_i$. For the pyrochlore model of Ref. [35], where the relaxation term vanishes by symmetry (see below), Eq. (39) reduces at $T = 0$ to the Goldstone-dispersion relation obtained there, including this factor of 8. The present formulation extends that relation to arbitrary $T < T_c$ and to sublattice-dependent pairing, for which the massive channels need not decouple and $\mathcal{Q}_{ij}$ carries the pair-geometric term of Eq. (31). The amplitude appearing in Eq. (39) is the squared pairing field in the normalization of Eq. (3); since $\mathcal{Q}_{ij}$ scales inversely under a rescaling of that field, only the product $|\Delta_0|^2 \mathcal{Q}_{ij}$ is independent of the choice.

The relaxation term vanishes precisely when $\langle \mathbf{w}_n | \partial_i \mathbf{M} | \mathbf{w}_0 \rangle = 0$ for all $n > 0$, equivalently $\mathbf{P}_{\perp} \partial_i \mathbf{w}_0 = \mathbf{0}$, in which case Eq. (39) reduces to the frozen-pairing expression (38), the uniform-pairing result of Refs. [3, 4]. This is a condition on the twisted problem, and it is not implied by equal gap magnitudes at $\mathbf{q} = \mathbf{0}$. Nor is it implied by time reversal, which maps $\Delta_0(\mathbf{q})$ onto $\Delta_0^*(-\mathbf{q})$ and therefore only requires $\mathbf{F}_i$ to be purely imaginary in a gauge where Δ_0 is real, i.e. the relative phases of the gaps to develop linearly in $\mathbf{q}$; nor by an inversion that permutes the orbitals. It does follow from an inversion that maps every orbital

onto itself, provided the orbitals are placed at their inversion-symmetric positions in the Fourier convention of Sec. II: $\Delta_0(\mathbf{q})$ and $\Delta_0(-\mathbf{q})$ then coincide up to a global phase, so that $\mathbf{F}_i$ is purely longitudinal and $\mathbf{P}_\perp \mathbf{F}_i = \mathbf{0}$. Like the two terms of Eq. (31) separately, the decoupling condition is thus a property of a given orbital embedding (Appendix B). It holds for the lattices of Refs. [35–37] and for the unstaggered Lieb lattice of Sec. VII at $\kappa = \frac{1}{2}$. At the other extreme, as a massive channel softens, $\lambda_n g_{00}^{nij}$ grows as λ_n^{-1} by Eq. (29), signaling the breakdown of the single-mode elimination rather than a physical divergence: that channel must then be kept with the Goldstone direction in an enlarged soft subspace. This happens as $T \rightarrow T_c^-$, treated in Sec. V.

V. CRITICAL REGIME

As $T \rightarrow T_c^-$ the amplitude mode of the critical channel becomes degenerate with the Goldstone direction at zero eigenvalue, so that the ordered-state kernel has a two-dimensional soft space; for the stationary zero-momentum critical point considered here the two do not mix at first order in $\mathbf{Q}$. As shown below, $\mathcal{Q}_{ij}$ has a finite critical limit, and what vanishes at the transition is the prefactor $|\Delta_0|^2$ of Eq. (39). Since the Goldstone-based construction has no counterpart above T_c , we work here with the normal-state pair propagator, to which the same soft-mode projection applies and yields the GL gradient coefficient directly.

A. GL gradient coefficient

At the normal-state saddle the global $U(1)$ symmetry is unbroken and the fluctuation field coincides with the pairing field, $\eta_S(Q) = \Phi_S(Q)$. The Gaussian fluctuation kernel then takes the block-diagonal form

$$\mathbf{M}(Q) = \begin{bmatrix} \Gamma^{-1}(Q) & 0 \\ 0 & [\Gamma^{-1}(-Q)]^T \end{bmatrix}, \quad (40)$$

where the $N_S \times N_S$ matrix $\Gamma^{-1}(Q)$ is the normal-state inverse pair propagator. The two blocks give identical contributions after relabeling $Q \rightarrow -Q$, so the quadratic effective action becomes $\mathcal{S}_{\text{eff}}^{(2)} = \sum_Q \boldsymbol{\eta}^\dagger(Q) \Gamma^{-1}(Q) \boldsymbol{\eta}(Q)$. As with the fluctuation kernel, we denote the static part of the inverse pair propagator by $\mathbf{K}(\mathbf{Q}) \equiv \Gamma^{-1}(\mathbf{Q}, i\nu_m = 0)$, which is Hermitian. We assume a zero-momentum transition with a single critical pairing channel and write

$$\mathbf{K}(\mathbf{0}; T) \mathbf{v}_n(T) = \lambda_n^{(N)}(T) \mathbf{v}_n(T), \quad (41)$$

with $\langle \mathbf{v}_m(T) | \mathbf{v}_n(T) \rangle = \delta_{mn}$. At T_c the Thouless criterion gives $\lambda_0^{(N)}(T_c) = 0$, while $\lambda_n^{(N)}(T_c) > 0$ for $n > 0$. In the following we write $\mathbf{v}_n \equiv \mathbf{v}_n(T_c)$ and evaluate $\lambda_n^{(N)}$ at

T_c unless the temperature argument is displayed; the superscript distinguishes these N_S normal-state eigenvalues from the stiffnesses λ_n of the doubled ordered-state kernel in Eq. (23). The critical eigenvector $\mathbf{v}_0$ determines the relative orbital structure of the pairing field at the transition.

Near T_c we decompose the pairing field into the critical mode and the remaining noncritical modes, $\boldsymbol{\eta}(\mathbf{Q}) = \Theta(\mathbf{Q}) \mathbf{v}_0 + \sum_{n>0} \rho_n(\mathbf{Q}) \mathbf{v}_n$, in analogy with Eq. (24) but in the undoubled channel space; the critical-mode amplitude $\Theta(\mathbf{Q})$ and the noncritical amplitudes $\rho_n(\mathbf{Q})$ are distinct from the Goldstone and massive amplitudes $\theta(\mathbf{Q})$ and $\chi_n(\mathbf{Q})$ of the ordered state. In this basis the static kernel takes the block form $\mathbf{K}(\mathbf{Q}) = \begin{bmatrix} K_{00}(\mathbf{Q}) & \mathbf{K}_{0\perp}(\mathbf{Q}) \\ \mathbf{K}_{\perp 0}(\mathbf{Q}) & \mathbf{K}_{\perp\perp}(\mathbf{Q}) \end{bmatrix}$, with blocks defined as in Eq. (25) with $\mathbf{w}_n \rightarrow \mathbf{v}_n$. Since $\lambda_n^{(N)} > 0$ for $n > 0$, $\mathbf{K}_{\perp\perp}(\mathbf{Q})$ is invertible at $\mathbf{Q} = \mathbf{0}$ and remains so for sufficiently small $\mathbf{Q}$, and the noncritical modes can be integrated out, giving the effective kernel of the critical mode,

$$K_{\text{eff}}(\mathbf{Q}) = K_{00}(\mathbf{Q}) - \mathbf{K}_{0\perp}(\mathbf{Q}) \mathbf{K}_{\perp\perp}^{-1}(\mathbf{Q}) \mathbf{K}_{\perp 0}(\mathbf{Q}), \quad (42)$$

the normal-state counterpart of Eq. (26).

We define $a(T) = K_{\text{eff}}(\mathbf{0}; T)$, with $a(T_c) = 0$ by the critical-mode condition. For a zero-momentum critical point, $\mathbf{Q} = \mathbf{0}$ is assumed to be a stationary point of the critical branch, so that $\partial_i K_{\text{eff}} = 0$. Below T_c , Eq. (22) and the doubled particle-hole structure imply the stronger condition that $M_{00}(\mathbf{Q})$ is even in $\mathbf{Q}$. No such evenness follows automatically in the undoubled normal-state problem, although it may be enforced, for example, by inversion or time-reversal symmetry. To leading order in the GL expansion, $K_{\text{eff}}(\mathbf{Q}; T) = a(T) + \frac{1}{2} \sum_{ij} c_{ij} Q_i Q_j + \dots$, with $c_{ij} = \partial_i \partial_j K_{\text{eff}}$. Expanding Eq. (42) to second order in $\mathbf{Q}$ gives

$$c_{ij} = \langle \mathbf{v}_0 | \partial_i \partial_j \mathbf{K} | \mathbf{v}_0 \rangle - 2 \sum_{n>0} \frac{\text{Re} [\langle \mathbf{v}_0 | \partial_i \mathbf{K} | \mathbf{v}_n \rangle \langle \mathbf{v}_n | \partial_j \mathbf{K} | \mathbf{v}_0 \rangle]}{\lambda_n^{(N)}}. \quad (43)$$

Differentiating the eigenvalue equation of the critical branch $\mathbf{v}_0(\mathbf{Q})$ of $\mathbf{K}(\mathbf{Q})$ at $\mathbf{Q} = \mathbf{0}$ and projecting onto a noncritical mode gives $\langle \mathbf{v}_n | \partial_i \mathbf{K} | \mathbf{v}_0 \rangle = -\lambda_n^{(N)} \langle \mathbf{v}_n | \partial_i \mathbf{v}_0 \rangle$, the analogue of Eq. (29); its derivation used only $\lambda_0^{(N)} = 0$ and the Hermiticity of the static kernel, both of which hold at T_c . Writing

$$g_{0\mathbf{Q}}^{(N)ij} = 2 \text{Re} [\langle \partial_{Q_i} \mathbf{v}_0(\mathbf{Q}) | \mathbf{v}_n(\mathbf{Q}) \rangle \langle \mathbf{v}_n(\mathbf{Q}) | \partial_{Q_j} \mathbf{v}_0(\mathbf{Q}) \rangle] \quad (44)$$

for the pair quantum metric of Eq. (30) built from the eigenvectors $\mathbf{v}_n(\mathbf{Q})$ of $\mathbf{K}(\mathbf{Q})$ rather than those of $\mathbf{M}(\mathbf{Q})$, the gradient coefficient becomes

$$c_{ij} = \langle \mathbf{v}_0 | \partial_i \partial_j \mathbf{K} | \mathbf{v}_0 \rangle - \sum_{n>0} \lambda_n^{(N)} g_{00}^{(N)ij}. \quad (45)$$

The GL gradient coefficient therefore carries the same stiffness-weighted pair-geometric correction as the

ordered-state stiffness: Eq. (45) has the structure of Eq. (31) with the ordered-state kernel replaced by the normal-state one and without the overall factor $\frac{1}{2}$, so that

$$c_{ij} = 2 \lim_{T \rightarrow T_c^-} \mathcal{Q}_{ij}(T). \quad (46)$$

As $T \rightarrow T_c^-$ the anomalous blocks of the ordered-state kernel vanish with the order parameter, so that $\mathbf{M}(\mathbf{Q})$ approaches the block-diagonal form (40) and the curvature of the Goldstone branch approaches that of the critical normal-state eigenvalue; this is the precise sense in which the two coefficients coincide [38]. In the remainder of this section $\mathcal{Q}_{ij}$ denotes this critical limit, $\mathcal{Q}_{ij} \equiv c_{ij}/2$.

The same limit can be taken term by term in Eq. (13). Below T_c the critical mode acquires a uniform equilibrium amplitude Θ_0 , with $\Delta_0 = \Theta_0 \mathbf{v}_0 + O(|\Theta_0|^3)$ and hence $|\Delta_0|^2 = |\Theta_0|^2 + O(|\Theta_0|^4)$ since $\mathbf{v}_0$ is normalized. For the block-diagonal kernel (40) the Goldstone vector is $\mathbf{w}_0 = (i\Theta_0 \mathbf{v}_0, -i\Theta_0^* \mathbf{v}_0^*)^T / (\sqrt{2}|\Theta_0|)$, and the two blocks contribute equally, so that the normalization factor $\frac{1}{2}$ cancels and $\langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle = \langle \mathbf{v}_0 | \partial_i \partial_j \mathbf{K} | \mathbf{v}_0 \rangle$. Equation (38) then gives $V[D_{\text{frozen}}]_{ij} = 4N_c |\Theta_0|^2 \langle \mathbf{v}_0 | \partial_i \partial_j \mathbf{K} | \mathbf{v}_0 \rangle + O(|\Theta_0|^4)$. In the same limit, each noncritical normal-state mode $\mathbf{v}_n$ generates two massive modes in the doubled space, one from each block of Eq. (40), both with stiffness $\lambda_n \rightarrow \lambda_n^{(N)}$. Because the Goldstone vector carries equal weight in the two blocks, each of these two modes contributes one half of $g_{00}^{(N)nij}$. Their sum therefore reproduces the corresponding undoubled normal-state contribution in Eq. (37), so that $\mathbf{F}_i^\dagger \mathbf{H}(\mathbf{0}) \mathbf{F}_j = 4|\Theta_0|^2 \sum_{n>0} \lambda_n^{(N)} g_{00}^{(N)nij} + O(|\Theta_0|^4)$. Combining the two contributions gives, to leading order near T_c ,

$$V[D_s]_{ij} = 4N_c |\Delta_0|^2 c_{ij}, \quad (47)$$

consistent with Eqs. (39) and (46).

For the uniform critical pairing channel of the pyrochlore model of Ref. [35], Eq. (47) reproduces the near- T_c relation obtained there; here the critical mode need not be uniform across the sublattices, and c_{ij} then carries the pair-geometric term of Eq. (45).

B. Coherence length from the GL functional

Projecting the static action onto the critical direction gives the GL free-energy density per unit cell,

$$f_{\text{GL}} = a(T)|\Theta|^2 + \frac{b_0}{2}|\Theta|^4 + \sum_{ij} \mathcal{Q}_{ij} \nabla_i \Theta^* \nabla_j \Theta, \quad (48)$$

where $\Theta(\mathbf{r})$ is the amplitude along $\mathbf{v}_0$, the real-space counterpart of $\Theta(\mathbf{Q})$ in the Fourier convention of Sec. III A, with $\nabla_i \leftrightarrow iQ_i$, so that the gradient term is the real-space form of the curvature $\sum_{ij} \mathcal{Q}_{ij} Q_i Q_j$ of $K_{\text{eff}}(\mathbf{Q})$, and $b_0 > 0$ is the coefficient of the quartic term generated

by expanding the $\text{Tr} \ln$ of Eq. (16) to fourth order in $\boldsymbol{\eta}$ and projecting onto $\mathbf{v}_0$. The noncritical channels do not correct b_0 at this order. At $\mathbf{Q} = \mathbf{0}$ and $T = T_c$ the off-diagonal blocks of Eq. (42) vanish, since $\mathbf{v}_0$ diagonalizes the static kernel there, so that $a(T) = \langle \mathbf{v}_0 | \mathbf{K}(\mathbf{0}; T) | \mathbf{v}_0 \rangle$ and this coincides with $\lambda_0^{(N)}(T)$ to first order in $T - T_c$, the drift of the critical eigenvector contributing only at second order. Writing $\varepsilon = (T_c - T)/T_c$ and linearizing,

$$a(T) \simeq -a_0 \varepsilon, \quad a_0 = T_c \partial_T \lambda_0^{(N)}|_{T_c} > 0, \quad (49)$$

minimization of Eq. (48) gives $|\Theta_0|^2 = a_0 \varepsilon / b_0$, so that the superfluid weight vanishes linearly at the transition, $V[D_s]_{ij} = 4N_c a_0 \varepsilon c_{ij} / b_0$, through the amplitude alone and not through the gradient coefficient. Inserting a uniform twist, $\Theta = |\Theta_0| e^{i\phi}$ with $\nabla_i \phi = 2q_i$, into the gradient term of Eq. (48) gives $V[D_s]_{ij} = 4N_c |\Theta_0|^2 c_{ij}$, which is Eq. (47) at the order to which it holds.

The coherence-length tensor follows from Eq. (48) without reference to any mass scale, as the ratio of the gradient and quadratic coefficients,

$$[\xi_{\text{GL}}^2]_{ij} = \frac{c_{ij}}{2a_0}, \quad [\xi^2(T)]_{ij} = [\xi_{\text{GL}}^2]_{ij} / \varepsilon. \quad (50)$$

Because c_{ij} carries the pair-geometric correction of Eq. (45), so does the coherence length. The correction enters with a negative sign and a_0 is unaffected by it, so the relaxation of the noncritical channels shortens every principal value of ξ_{GL} relative to the value obtained by holding the pairing configuration rigid across the sublattices. This is the same softening that gives $\mathbf{D}_s \preceq \mathbf{D}_{\text{frozen}}$ below T_c : both quantities are proportional to the relaxed curvature $\mathcal{Q}_{ij}$. When a single length is wanted in place of the tensor, the natural choice is the geometric mean of its principal-axis values, $\bar{\xi}_{\text{GL}} = [\det \xi_{\text{GL}}^2]^{\frac{1}{2d}}$, where $\bar{d}$ is the number of spatial dimensions in which the system is extended; $\bar{\xi}_{\text{GL}}$ is the length whose $\bar{d}$ -dimensional volume matches that of the coherence ellipsoid. Quantum-geometric contributions to the coherence length of flat-band superconductors were discussed in the uniform-pairing setting [39–41].

C. Pair mass from the time-dependent GL theory

The construction so far has used only the static kernel. To exhibit the dynamics, we retain the frequency dependence of the normal-state propagator, $\Gamma^{-1}(\mathbf{Q}, i\nu_m)$, continue analytically to real frequency, $i\nu_m \rightarrow \omega + i0^+$, and extend the kernel accordingly, $\mathbf{K}(\mathbf{Q}, \omega) \equiv \Gamma^{-1}(\mathbf{Q}, \omega + i0^+)$, so that $\mathbf{K}(\mathbf{Q}, 0) = \mathbf{K}(\mathbf{Q})$. The dynamical coefficient of the critical mode is

$$d = -\partial_\omega \langle \mathbf{v}_0 | \mathbf{K}(\mathbf{0}, \omega) | \mathbf{v}_0 \rangle \big|_{\omega=0, T_c}. \quad (51)$$

The noncritical channels do not correct d . At $\mathbf{Q} = \mathbf{0}$ the mixed block $\mathbf{K}_{\perp 0}(\mathbf{0}, \omega)$ vanishes at $\omega = 0$, because $\mathbf{v}_0$ diagonalizes the static kernel; its leading frequency dependence is therefore linear in ω , and the Schur-complement

correction in Eq. (42) begins at order ω^2 . The pair-geometric correction thus resides entirely in the gradient coefficient and not in the dynamical one. Collecting the three coefficients, the effective kernel of the critical mode has the time-dependent GL form

$$K_{\text{eff}}(\mathbf{Q}, \omega) = a(T) - d\omega + \sum_{ij} \mathcal{Q}_{ij} Q_i Q_j + \dots, \quad (52)$$

whose coefficients $a(T)$, d and $\mathcal{Q}_{ij}$ are all defined directly from the pair kernel.

Equations (39), (45) and (50) require no propagating pair mode, and do not depend on the interpretation that follows. When the low-energy pair mode is well enough defined that d is real and positive, as it is when the mode lies below the two-particle continuum, the canonically normalized field $\Theta_B = \sqrt{d}\Theta$ obeys a Gross-Pitaevskii equation whose inverse pair-mass tensor, pair repulsion, chemical potential and condensate density are [39, 42]

$$\begin{aligned} [m_B^{-1}]_{ij} &= \frac{c_{ij}}{d}, & U_{\text{BB}} &= \frac{b_0}{d^2}, \\ \mu_B &= \frac{a_0 \varepsilon}{d}, & n_{\text{pair}}^c &= \frac{N_c d |\Theta_0|^2}{V}. \end{aligned} \quad (53)$$

The quartic coefficient b_0 of Eq. (48) thus appears as a repulsive contact interaction between pairs, and the Gross-Pitaevskii equation is completely specified by the coefficients of Eqs. (48) and (52). The identification is internally consistent: at the equilibrium amplitude $|\Theta_{B,0}|^2 = d|\Theta_0|^2 = da_0\varepsilon/b_0$ one finds $U_{\text{BB}}|\Theta_{B,0}|^2 = a_0\varepsilon/d = \mu_B$, as a weakly interacting Bose condensate requires. For a single band this reduces to the pair mass of Ref. [42]. In this regime, and only in this regime, Eq. (47) takes the Bose form $V[D_s]_{ij} = 4Vn_{\text{pair}}^c[m_B^{-1}]_{ij}$ near T_c , the factor of 4 being the pair charge squared [35, 36]. In the dilute limit, where $\Gamma^{-1}(\mathbf{Q})$ reduces to the inverse vacuum two-body $\mathcal{T}$ matrix, $[m_B^{-1}]_{ij} = c_{ij}/d$ coincides with the exact effective mass of the two-body bound state on a multiband lattice obtained in Ref. [28], including its pair-geometric contribution.

In the same reading, the coherence length of Eq. (50) is the healing length of this Bose gas [43]. Writing the standard Gross-Pitaevskii relation $[\xi_B^2(T)]_{ij} = [m_B^{-1}]_{ij}/2\mu_B$ in terms of its temperature-independent amplitude, $[\xi_B^2(T)]_{ij} = [\xi_B^2]_{ij}/\varepsilon$, gives

$$[\xi_B^2]_{ij} = \varepsilon \frac{[m_B^{-1}]_{ij}}{2\mu_B} = \frac{c_{ij}}{2a_0}, \quad (54)$$

which is the first of Eqs. (50). The dynamical coefficient cancels between the inverse mass and the chemical potential, which is why the coherence length follows from the static kernel alone while the pair mass does not: only the ratio c_{ij}/a_0 is observable in ξ_B , however heavy the pair.

Two remarks attach to Eq. (53). First, n_{pair}^c is not a condensate density in the sense of off-diagonal long-range order but the normalization that the Gross-Pitaevskii

form imposes on the pairing field; the canonical factor $\sqrt{d}$ is what makes the entries of Eq. (53) individually independent of the Hubbard-Stratonovich normalization, unlike the product fixed by Eq. (39). Second, when low-energy quasiparticles are available at the energy of the critical mode, d acquires an imaginary part describing its damping [42], and the pair mass loses its meaning.

VI. VALIDITY AND SCOPE

Our derivation rests on the following assumptions: (i) the mean-field saddle-point approximation and the Gaussian expansion of the pairing-field action about it; (ii) a uniform saddle point, which in the twisted case means uniform after the gauge transformation of Sec. II; (iii) local stability against static fluctuations at finite momentum, $\mathbf{M}(\mathbf{Q}) \succeq 0$ in a neighborhood of $\mathbf{Q} = \mathbf{0}$; (iv) a single soft static mode below T_c , namely the Goldstone direction, with all other static eigenvalues positive; and (v) analyticity of $\mathbf{M}(\mathbf{Q})$ about $\mathbf{Q} = \mathbf{0}$ with a nondegenerate Goldstone branch, so that $\mathbf{w}_0(\mathbf{Q})$ can be continued smoothly. No time-reversal or inversion symmetry is assumed: below T_c the particle-hole structure (22) supplies the evenness of the Goldstone projection, and at T_c only the stationarity of the critical branch at $\mathbf{Q} = \mathbf{0}$ is required (Sec. V A). Within these assumptions the channel structure is arbitrary. Finite-range and interorbital pairing, spin-mixing spin-orbit coupling, complex hoppings and coexisting mean fields only enlarge the set of pairing channels; and if several continuous symmetries are broken simultaneously, the projection onto $\mathbf{w}_0$ is replaced by a projection onto the soft subspace.

Note that $\mathcal{Q}_{ij}$ is defined from the static kernel alone (Sec. V C). In a system with gapless excitations the dynamical kernel $\mathbf{M}(\mathbf{Q}, i\nu_m)$, after analytic continuation, contains quasiparticle damping [42], and the limits $\omega \rightarrow 0$ and $\mathbf{Q} \rightarrow \mathbf{0}$ need not commute; the thermodynamic superfluid weight is the static response, with the zero-frequency limit taken first. Gapless fermions may also produce a nonanalytic momentum dependence of the static kernel, which has to be checked case by case.

Assumption (i) is more basic. Beyond the Gaussian level the identification $\mathbf{M}(\mathbf{0}) = \mathcal{H}(\mathbf{0})$ and the closed form in terms of the mean-field amplitude are no longer available, although the eigenvalue route of Eq. (28) can still be applied to the static pair kernel of whatever approximation is used.

VII. NUMERICAL BENCHMARK

We now benchmark numerically the central identities of Secs. II–IV for the two-dimensional Lieb lattice at $T = 0$ [24]: the thermodynamic relaxation term of Eq. (13) against its pair-metric form, Eq. (37), and the superfluid weight of Eq. (13) against Eqs. (39) and (28). All three are relations at fixed temperature, so a benchmark at $T =$

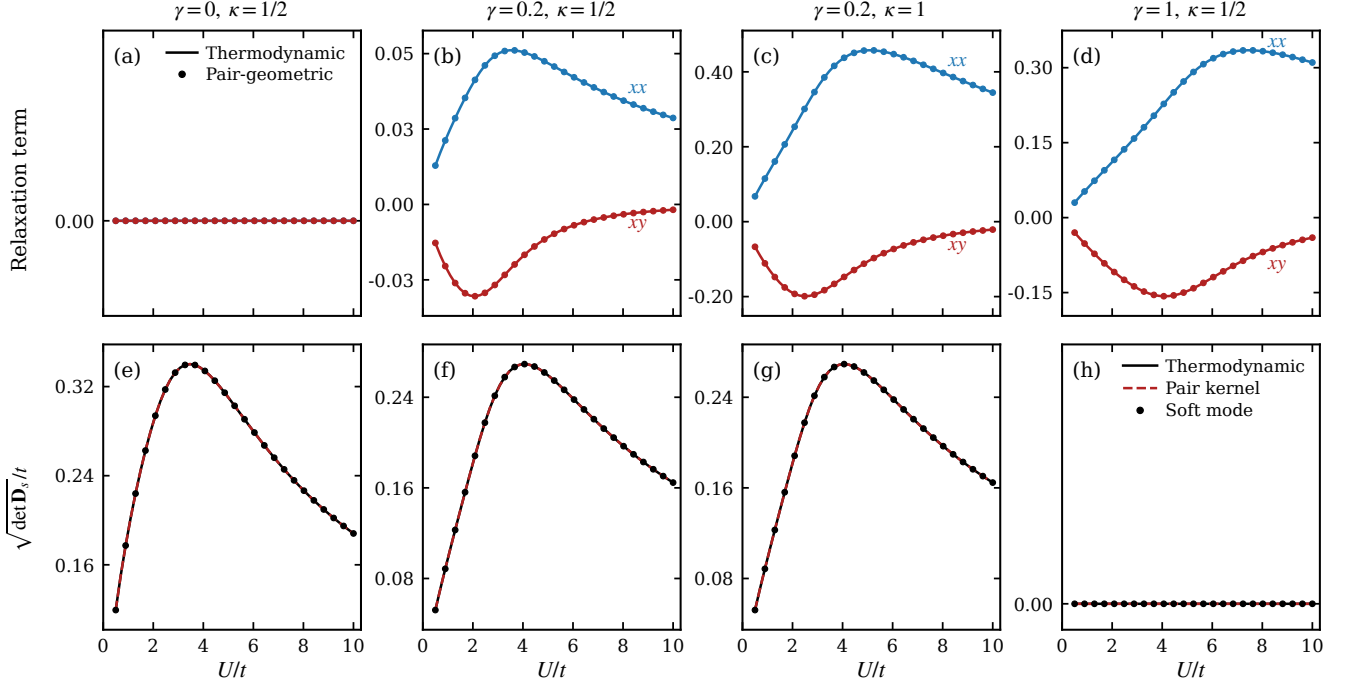


FIG. 1. Numerical benchmark at $T = 0$ as a function of the interaction strength U/t . The upper row, panels (a)–(d), compares the thermodynamic and pair-geometric representations of the relaxation term, shown in units of t . Solid lines show the components of $\mathbf{F}_i^\dagger \mathbf{H}(\mathbf{0}) \mathbf{F}_j$ obtained from the self-consistent gap response, while filled circles show the stiffness-weighted pair quantum metric of Eq. (37). Blue and red denote the xx and xy components, respectively. The lower row, panels (e)–(h), shows $\sqrt{\det \mathbf{D}_s}/t$ obtained from the thermodynamic formulation (black solid lines), the pair-fluctuation-kernel formulation (red dashed lines), and the curvature of $\lambda_0(\mathbf{Q})$ (black filled circles) according to Eq. (28). The columns correspond to (a,e) $(\gamma, \kappa) = (0, \frac{1}{2})$, (b,f) $(\gamma, \kappa) = (0.2, \frac{1}{2})$, (c,g) $(\gamma, \kappa) = (0.2, 1)$, and (d,h) $(\gamma, \kappa) = (1, \frac{1}{2})$.

0 tests them completely; what the temperature controls is the value of $\mathcal{Q}_{ij}$ and of the prefactor, not the identities themselves.

The Lieb lattice has three orbitals $S = A, B, C$ per unit cell, at $\mathbf{x}_A = (0, \kappa)$, $\mathbf{x}_B = (0, 0)$ and $\mathbf{x}_C = (\kappa, 0)$ in units of the lattice constant, which we set to unity, where κ parametrizes the orbital embedding and $\kappa = \frac{1}{2}$ corresponds to the geometric positions of the orbitals. The nearest-neighbor hopping amplitudes are staggered as $t_\pm = t(1 \pm \gamma)$, with t_+ within the unit cell and t_- between neighboring cells. In the orbital basis (A, B, C) , the corresponding Bloch Hamiltonian $\mathbf{h}_{\mathbf{k}\uparrow} = \mathbf{h}_{\mathbf{k}\downarrow}$ is

$$\mathbf{h}_{\mathbf{k}} = \begin{bmatrix} 0 & f_y(\mathbf{k}) & 0 \\ f_y^*(\mathbf{k}) & 0 & f_x^*(\mathbf{k}) \\ 0 & f_x(\mathbf{k}) & 0 \end{bmatrix}, \quad (55)$$

where $f_x(\mathbf{k}) = t_+ e^{-ik_x \kappa} + t_- e^{ik_x(1-\kappa)}$ and $f_y(\mathbf{k}) = t_+ e^{-ik_y \kappa} + t_- e^{ik_y(1-\kappa)}$. Thus γ controls the hopping staggering, while κ changes the embedding without changing the single-particle spectrum, and is therefore the parameter against which the invariance of Eq. (39) discussed in Appendix B can be tested directly. For each U/t the pairing amplitudes of Eq. (3) are determined self-consistently from Eq. (7). The mirror symmetry $x \leftrightarrow y$ through the B site relates the A and C orbitals, so that $\Delta_{0A} = \Delta_{0C}$ and

the zero-twist pairing vector is $\Delta_0 = (\Delta_{0A}, \Delta_{0B}, \Delta_{0A})^T$. We work over $0.5 \leq U/t \leq 10$ for the four parameter sets $(\gamma, \kappa) = (0, \frac{1}{2})$, $(0.2, \frac{1}{2})$, $(0.2, 1)$ and $(1, \frac{1}{2})$. The first is a high-symmetry reference point; the two cases at $\gamma = 0.2$ probe the dependence of the separate contributions on the orbital embedding at fixed hopping staggering; and $\gamma = 1$ is the disconnected limit, which provides a parameter-free check on the complete superfluid weight. These are the parameter sets of the Lieb-lattice benchmark of Ref. [24].

The upper row of Fig. 1 compares the two representations of the relaxation term. As discussed in Sec. II, the thermodynamic contribution $\mathbf{F}_i^\dagger \mathbf{H}(\mathbf{0}) \mathbf{F}_j$ is the order-parameter relaxation term identified in Ref. [24], obtained there by solving the differentiated gap equation, whereas Eq. (37) expresses the same quantity as a sum over massive channels weighted by their static stiffnesses, with no reference to the gap response at all. The two evaluations agree to numerical precision across the whole interaction range for all four parameter sets. At the high-symmetry point $(\gamma, \kappa) = (0, \frac{1}{2})$ the relaxation term vanishes, while for finite staggering it is nonzero and depends strongly on the embedding. It is worth noting that the gaps are unequal, $\Delta_{0A} \neq \Delta_{0B}$, in every panel, including the one in which the relaxation vanishes: equal gap mag-

nitudes at zero twist are not needed for the decoupling condition of Sec. IV, which at $(\gamma, \kappa) = (0, \frac{1}{2})$ follows instead from the inversion symmetry about the B site; for $\gamma = 0$ but $\kappa \neq \frac{1}{2}$ the relaxation term is nonzero [24].

The lower row of Fig. 1 shows $\sqrt{\det \mathbf{D}_s}$, the geometric mean of the principal values of the superfluid-weight tensor, obtained in the three ways. The three agree to numerical precision throughout. Two features deserve comment. First, although the relaxation contribution differs substantially between the embeddings $\kappa = \frac{1}{2}$ and $\kappa = 1$ at fixed $\gamma = 0.2$, as panels (b) and (c) of Fig. 1 show, the superfluid weight is identical in the two cases. This is the embedding independence emphasized in Ref. [24], and here it appears as a consequence of Eq. (28): the direct curvature and the pair-geometric term are separately embedding dependent, while the eigenvalue whose curvature they combine to give is not (Appendix B). The compensation that must be arranged term by term in the thermodynamic formulation is automatic in the collective-fluctuation one. Second, the disconnected limit $\gamma = 1$ provides a check that requires no comparison between methods. There $t_- = 0$, the unit cells decouple, and no current can flow, so the superfluid weight must vanish identically at every U/t . The frozen-pairing contribution does not vanish: holding the pairing configuration rigid under the twist assigns a finite superfluid weight to a system of disconnected clusters. It is cancelled exactly, at every interaction strength, by the stiffness-weighted pair quantum metric of Eq. (37), and panel (h) of Fig. 1 shows the cancellation to numerical precision. This is the most direct demonstration that the relaxation term is not a small correction to be estimated but a necessary part of the superfluid weight [24].

VIII. CONCLUSION

We showed that the thermodynamic formulation of the superfluid weight, in which the self-consistent relaxation of the pairing amplitudes enters through the Hessian of the thermodynamic potential [24], has an equivalent description in terms of collective pairing fluctuations: the Hessian is the static pair-fluctuation kernel at zero momentum, the relaxation correction is the Schur complement of the massive pairing channels at finite pair momentum, and the superfluid weight is the small-momentum curvature of the Goldstone branch of the kernel.

This reformulation exposed a quantum-geometric structure. As the pair momentum is varied, the Goldstone eigenvector traces a path through the space of pairing configurations, and the relaxation correction is its quantum metric weighted by the static stiffnesses of the massive channels, a collective counterpart of the Bloch quantum metric of the band structure. The same geometry enters the GL gradient coefficient and the coherence-length tensor near T_c , and, once the frequency dependence of the kernel is retained, the pair mass [28]. The

Lieb-lattice benchmark confirmed these identities and the embedding independence of the total superfluid weight, including the exact cancellation of the spurious frozen-pairing stiffness in the disconnected limit [24].

Natural directions for future work include systems with several soft collective modes, and the role of the pair Berry curvature and of the topology of the normal-state pair-fluctuation branches when time-reversal symmetry is broken.

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Appendix A: Alternative derivation of Eq. (20)

Here we derive Eq. (20) directly from the effective action of Eq. (16), by evaluating the thermodynamic Hessian $\mathcal{H}(\mathbf{0})$ of Eq. (10) and the Gaussian fluctuation kernel $\mathbf{M}(Q)$ of Eq. (19) as second derivatives of $\mathcal{S}_{\text{eff}}$ from the same starting point. It is convenient to split the action as $\mathcal{S}_{\text{eff}} = \mathcal{S}_\Phi + \mathcal{S}_f$, where $\mathcal{S}_\Phi = \frac{1}{V} \sum_{QS} |\Phi_S(Q)|^2$ and $\mathcal{S}_f = -\text{Tr} \ln [-\mathbf{G}^{-1}]$, with $\mathbf{G}^{-1}[\Phi]$ given by Eq. (15).

The BdG Hamiltonian is linear in the doubled pairing vector $\Psi = (\Delta, \Delta^*)^T$, $\mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{0}) = \mathbf{h}_{\text{BdG}}^{(N)}(\mathbf{k}) + \sum_{\alpha=1}^{2N_S} \Psi_\alpha \mathbf{V}_\alpha$, with $\mathbf{V}_\alpha = \frac{\partial \mathbf{h}_{\text{BdG}}}{\partial \Psi_\alpha}$, where $\mathbf{h}_{\text{BdG}}^{(N)}(\mathbf{k})$ is the normal-state BdG matrix of Eq. (15). Labeling the first N_S Nambu-orbital states as particle-like and the last N_S as hole-like, on-site pairing gives $\mathbf{V}_S = |S\rangle\langle N_S+S|$ and $\mathbf{V}_{N_S+S} = \mathbf{V}_S^\dagger$ for $S = 1, \dots, N_S$: the pairing vertices connect the particle and hole components associated with the same orbital S , and they carry no $\mathbf{k}$ dependence. The doubling of the pairing vector implies $\Psi_\alpha^* = \Psi_{\bar{\alpha}}$ with $\bar{\alpha} = \alpha + N_S \pmod{2N_S}$, and therefore also $\mathbf{V}_\alpha^* = \mathbf{V}_{\bar{\alpha}}$ and $\partial \Psi_\alpha^* = \partial \Psi_{\bar{\alpha}}$.

For a static and spatially uniform field, Eq. (15) is diagonal in k with blocks $i\omega_n \mathbf{1} - \mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{0})$, so that $\partial \mathbf{G}^{-1} / \partial \Psi_\alpha = -\mathbf{V}_\alpha$. Using $\delta \text{Tr} \ln \mathbf{A} = \text{Tr}(\mathbf{A}^{-1} \delta \mathbf{A})$ with $\mathbf{A} = -\mathbf{G}^{-1}$, and then $\partial \mathbf{G} = -\mathbf{G}(\partial \mathbf{G}^{-1})\mathbf{G}$, gives $\frac{\partial \mathcal{S}_f}{\partial \Psi_\alpha} = \text{Tr}[\mathbf{G} \mathbf{V}_\alpha]$, and $\frac{\partial^2 \mathcal{S}_f}{\partial \Psi_\alpha^* \partial \Psi_{\alpha'}} = \text{Tr}[\mathbf{G} \mathbf{V}_\alpha^\dagger \mathbf{G} \mathbf{V}_{\alpha'}]$, where the second follows from the first by differentiating once more and using $\partial \Psi_\alpha^* = \partial \Psi_{\bar{\alpha}}$ together with $\mathbf{V}_{\bar{\alpha}} = \mathbf{V}_\alpha^\dagger$. The pairing contribution to the same derivative is immediate: for a uniform static configuration with arbitrary Ψ , $\Phi_S(Q) = \sqrt{\beta N_c} \Delta_S \delta_{Q0}$, so that $\mathcal{S}_\Phi = \beta N_c \sum_S |\Delta_S|^2 / U$, and since Δ and Δ^* are independent variables, $(\beta N_c)^{-1} \partial \Psi_\alpha^* \partial \Psi_{\alpha'} \mathcal{S}_\Phi = \delta_{\alpha\alpha'} / U$. Because $\mathcal{S}_{\text{eff}} = \beta N_c \Omega_0(\mathbf{0}, \Psi)$ for such configurations, the Hessian of Eq. (10) is obtained by dividing the sum of the two contributions by βN_c and evaluating the result at the

saddle $\Psi = \Psi_0$,

$$[\mathcal{H}(\mathbf{0})]_{\alpha\alpha'} = \frac{\delta_{\alpha\alpha'}}{U} + \frac{1}{\beta N_c} \sum_k \text{tr} [\mathbf{G}_0(k) \mathbf{V}_\alpha^\dagger \mathbf{G}_0(k) \mathbf{V}_\alpha], \quad (\text{A1})$$

where $\mathbf{G}_0(k) = [i\omega_n \mathbf{1} - \mathbf{h}_{\text{BdG}}(\mathbf{k}, \mathbf{0})]^{-1}$ is the Nambu Green's function at the self-consistent saddle.

We next obtain $\mathbf{M}(Q)$ from the same action. Writing $\mathbf{G}^{-1} = \mathbf{G}_0^{-1} + \delta\mathbf{G}^{-1}$ and expanding the logarithm, the quadratic fermionic contribution is $\mathcal{S}_f^{(2)} = \frac{1}{2} \text{Tr}[\mathbf{G}_0 \delta\mathbf{G}^{-1} \mathbf{G}_0 \delta\mathbf{G}^{-1}]$. The factor $\frac{1}{2}$ here comes from the expansion of $\ln(1+x)$ and not from any spinor redundancy: every Grassmann field appears exactly once in the minimal $2N_S$ -component spinor $\mathbf{C}_k$, so that the fermionic determinant gives $-\text{Tr} \ln[-\mathbf{G}^{-1}]$ with unit coefficient. By Eq. (15), a fluctuation of four-momentum Q enters as $\delta\mathbf{G}^{-1} = -\frac{1}{\sqrt{\beta N_c}} \sum_{Q\alpha} \delta\Phi_\alpha(Q) \mathbf{V}_\alpha$, with $\delta\Phi(Q)$ the doubled fluctuation field of Sec. IIIB, and connects two propagators whose four-momenta differ by Q . The pairing contribution quadratic in the fluctuations is $\mathcal{S}_\Phi^{(2)} = U^{-1} \sum_{Q,S} |\eta_S(Q)|^2 = \frac{1}{2} \sum_Q \delta\Phi^\dagger(Q) (U^{-1} \mathbf{1}) \delta\Phi(Q)$, where the last equality uses the doubling constraint of Eq. (18); hence $\mathbf{M}^\Phi = U^{-1} \mathbf{1} = \mathcal{H}^\Phi$, and the two constructions already agree in this sector. Comparing the fermionic part with Eq. (19) then gives

$$[M(Q)]_{\alpha\alpha'} = \frac{\delta_{\alpha\alpha'}}{U} + \frac{1}{\beta N_c} \sum_k \text{tr} [\mathbf{G}_0(k) \mathbf{V}_\alpha^\dagger \mathbf{G}_0(k+Q) \mathbf{V}_{\alpha'}]. \quad (\text{A2})$$

Equations (A1) and (A2) have the same structure, and differ only through the argument of the second propagator. At $Q = (\mathbf{0}, 0)$ the two Green's functions carry the same four-momentum, Eq. (A2) reduces to Eq. (A1), and Eq. (20) follows; Eq. (21) is the static limit of Eq. (A2) after the Matsubara sum.

Appendix B: Orbital-embedding dependence

Here we show how a change of orbital embedding representing the same real-space Hamiltonian is implemented in momentum space, how it transforms the Bloch Hamiltonian and the static fluctuation kernel, and why the soft-mode curvature $\mathcal{Q}_{ij}$ is invariant under it while the two contributions of Eq. (31) separately are not. The embedding phases below depend on the pair momentum alone, so the Matsubara frequency plays no role and we work throughout with the static kernel $\mathbf{M}(\mathbf{Q})$ of Sec. IIIB.

We consider a change of orbital positions $\mathbf{x}_S \rightarrow \mathbf{x}_S + \mathbf{d}_S$ at fixed real-space hopping amplitudes, so that $\mathbf{r}_{iS} \rightarrow \mathbf{r}_{iS} + \mathbf{d}_S$. The real-space operators $c_{iS\sigma}$ are unchanged; what changes is the Fourier convention of Sec. II, which carries the orbital positions explicitly. Equating the two expansions of the same operator, $\sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_{iS}} c_{\mathbf{k}S\sigma} = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_{iS} + \mathbf{d}_S)} c'_{\mathbf{k}S\sigma}$, gives $c'_{\mathbf{k}S\sigma} = e^{-i\mathbf{k} \cdot \mathbf{d}_S} c_{\mathbf{k}S\sigma}$. Introducing the diagonal unitary matrix

$[W_{\mathbf{d}}(\mathbf{k})]_{SS'} = e^{-i\mathbf{k} \cdot \mathbf{d}_S} \delta_{SS'}$, this reads $\mathbf{c}'_{\mathbf{k}\sigma} = \mathbf{W}_{\mathbf{d}}(\mathbf{k}) \mathbf{c}_{\mathbf{k}\sigma}$, and the Bloch Hamiltonian therefore transforms as

$$\mathbf{h}'_{\mathbf{k}\sigma} = \mathbf{W}_{\mathbf{d}}(\mathbf{k}) \mathbf{h}_{\mathbf{k}\sigma} \mathbf{W}_{\mathbf{d}}^\dagger(\mathbf{k}), \quad (\text{B1})$$

so that its spectrum is unchanged. The on-site pair operator $b_{S\mathbf{Q}}^\dagger = \sum_{\mathbf{k}} c_{\mathbf{k}+\mathbf{Q},S\uparrow}^\dagger c_{-\mathbf{k},S\downarrow}^\dagger$ of Eq. (2) picks up the phases $e^{i(\mathbf{k}+\mathbf{Q}) \cdot \mathbf{d}_S}$ and $e^{-i\mathbf{k} \cdot \mathbf{d}_S}$ from its two factors. The $\mathbf{k}$ dependence cancels, as it does in Sec. II, leaving $b_{S\mathbf{Q}}'^\dagger = e^{i\mathbf{Q} \cdot \mathbf{d}_S} b_{S\mathbf{Q}}^\dagger$ and $b_{S\mathbf{Q}}' = e^{-i\mathbf{Q} \cdot \mathbf{d}_S} b_{S\mathbf{Q}}$. Hence $b_{S\mathbf{Q}}'^\dagger b_{S\mathbf{Q}}' = b_{S\mathbf{Q}}^\dagger b_{S\mathbf{Q}}$ and the interaction term is invariant. The embedding therefore changes the momentum-space representation of the single-particle sector and the phase convention of the pair operators, while leaving the full Hubbard Hamiltonian of Eq. (1) unchanged.

The Hubbard-Stratonovich field of Sec. IIIB, being conjugate to $b_S(Q)$, transforms oppositely so as to preserve that coupling,

$$\Phi_S'(Q) = e^{-i\mathbf{Q} \cdot \mathbf{d}_S} \Phi_S(Q), \quad (\text{B2})$$

so that in particular the uniform $\mathbf{Q} = \mathbf{0}$ pairing field, and with it the saddle Δ_0 of Eq. (3), is unchanged. For the doubled fluctuation field $\delta\Phi(\mathbf{Q}) = (\eta(\mathbf{Q}), \eta^*(-\mathbf{Q}))^T$ of Sec. IIIB both blocks transform with the same phase, since Eq. (B2) gives $\eta_S'^*(-\mathbf{Q}) = e^{-i\mathbf{Q} \cdot \mathbf{d}_S} \eta_S^*(-\mathbf{Q})$. Therefore $\delta\Phi'(\mathbf{Q}) = \mathcal{W}_{\mathbf{d}}(\mathbf{Q}) \delta\Phi(\mathbf{Q})$ with $\mathcal{W}_{\mathbf{d}}(\mathbf{Q}) = \text{diag}[\mathbf{W}_{\mathbf{d}}(\mathbf{Q}), \mathbf{W}_{\mathbf{d}}(\mathbf{Q})]$. Since $\mathcal{W}_{\mathbf{d}}(\mathbf{Q})$ is unitary and the quadratic action (19) describes the same physical configuration in either embedding, the corresponding kernels satisfy

$$\mathbf{M}'(\mathbf{Q}) = \mathcal{W}_{\mathbf{d}}(\mathbf{Q}) \mathbf{M}(\mathbf{Q}) \mathcal{W}_{\mathbf{d}}^\dagger(\mathbf{Q}), \quad (\text{B3})$$

a unitary similarity transformation at every $\mathbf{Q}$. Let $\mathbf{M}(\mathbf{Q}) \mathbf{w}_n(\mathbf{Q}) = \lambda_n(\mathbf{Q}) \mathbf{w}_n(\mathbf{Q})$, $n \geq 0$, define the eigenvalue branches of the static kernel, with $n=0$ the Goldstone branch of Sec. IIID. Equation (B3) implies that $\mathcal{W}_{\mathbf{d}}(\mathbf{Q}) \mathbf{w}_n(\mathbf{Q})$ is an eigenvector of $\mathbf{M}'(\mathbf{Q})$ with the same eigenvalue, so that $\lambda'_n(\mathbf{Q}) = \lambda_n(\mathbf{Q})$ for every branch and every $\mathbf{Q}$. In particular $\lambda'_0(\mathbf{Q}) = \lambda_0(\mathbf{Q})$, and Eq. (28) then gives

$$\mathcal{Q}'_{ij} = \frac{1}{2} \partial_i \partial_j \lambda'_0 = \mathcal{Q}_{ij}. \quad (\text{B4})$$

The two terms of Eq. (31) are not individually embedding invariant. To see this, write $\mathcal{W}_{\mathbf{d}}(\mathbf{Q}) = \exp(-i \sum_i Q_i \mathcal{X}_i)$ with $\mathcal{X}_i = \text{diag}(d_{1i}, \dots, d_{N_S i}, d_{1i}, \dots, d_{N_S i})$, where d_{Si} is the i th Cartesian component of $\mathbf{d}_S$. Thus $\mathcal{X}_i$ is the orbital-position operator in the doubled pairing channels, and a change of embedding is the $\mathbf{Q}$ -dependent gauge transformation it generates, i.e. precisely the kind of transformation under which a quantum metric is not invariant. Since $\mathcal{W}_{\mathbf{d}}(\mathbf{0}) = \mathbf{1}$, the zero-momentum kernel and its eigenvectors are untouched: $\mathbf{M}'(\mathbf{0}) = \mathbf{M}(\mathbf{0})$, $|\mathbf{w}'_n\rangle = |\mathbf{w}_n\rangle$ and $\lambda'_n = \lambda_n$. Differentiating Eq. (B3) at $\mathbf{Q} = \mathbf{0}$ gives

$$\partial_i \mathbf{M}' = \partial_i \mathbf{M} - i [\mathcal{X}_i, \mathbf{M}(\mathbf{0})], \quad (\text{B5})$$

while the second derivative acquires further terms involving $\mathcal{X}_i$ and $\mathcal{X}_j$, so that the direct-curvature term $\langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle$ changes. The pair-geometric term changes correspondingly: at finite momentum $|\mathbf{w}'_n(\mathbf{Q})\rangle = \mathcal{W}_d(\mathbf{Q})|\mathbf{w}_n(\mathbf{Q})\rangle$ up to a $\mathbf{Q}$ -dependent phase, which drops out of Eq. (30), and differentiating at $\mathbf{Q} = \mathbf{0}$ gives

$$|\partial_i \mathbf{w}'_n\rangle = |\partial_i \mathbf{w}_n\rangle - i\mathcal{X}_i |\mathbf{w}_n\rangle, \quad (\text{B6})$$

so that the mode-resolved metric g_{00}^{nij} of Eq. (30) is embedding dependent as well. The added term is not a pure phase, so neither quantity is invariant on its own; only their combination in Eq. (31), which equals $\frac{1}{2}\partial_i \partial_j \lambda_0$, is, by Eq. (B4). The Lieb-lattice benchmark of Sec. VII shows both statements.

Appendix C: Derivations of Eqs. (34) and (38)

Here we derive the relations between the phase-twist derivatives of the thermodynamic formulation of Sec. II and the momentum dependence of the static fluctuation kernel of Sec. III B: the mixed derivative $\mathbf{B}_i$ of Eq. (34) and the frozen-pairing curvature of Eq. (38). We write $\mathbf{r}$ for the position of the pairing component under consideration, with Cartesian components r_i , and we display all sums over repeated Cartesian indices. With $\mathbf{Q} = 2\mathbf{q}$ as in Sec. IV, every explicit $\mathbf{q}$ derivative at fixed Ψ_0 is a derivative along the winding configuration (32).

At zero twist an infinitesimal global phase rotation of the uniform saddle generates the Goldstone vector of Eq. (12),

$$\mathbf{u}_0 = i\tau_3 \Psi_0 = \sqrt{2}|\Delta_0| \mathbf{w}_0, \quad (\text{C1})$$

with $\mathbf{M}(\mathbf{0})\mathbf{u}_0 = \mathbf{0}$. To linear order in the twist the phase-twisted configuration is $\mathbf{U}_q(\mathbf{r})\Psi_0 = \Psi_0 + 2\sum_i q_i r_i \mathbf{u}_0 + O(q^2)$. Let $\mathbf{M}[\cdot]$ denote the static kernel as a real-space operator, i.e. the operator whose momentum representation is $\mathbf{M}(\mathbf{Q})$, with square brackets indicating the field it acts on; since it is the Hessian of the static effective action at the saddle, the gradient generated by a small static variation $\delta\Phi$ is $\mathbf{M}[\delta\Phi]$ to linear order. The coefficient of q_i in the gradient on the twisted configuration is therefore $\mathbf{B}_i = 2\mathbf{M}[r_i \mathbf{u}_0]$. To express this in momentum space we use $r_i \mathbf{u}_0 = -i\partial_{Q_i}[e^{i\mathbf{Q}\cdot\mathbf{r}}\mathbf{u}_0]_{\mathbf{Q}=\mathbf{0}}$ together with translational invariance, $\mathbf{M}[e^{i\mathbf{Q}\cdot\mathbf{r}}\mathbf{u}_0] = e^{i\mathbf{Q}\cdot\mathbf{r}}\mathbf{M}(\mathbf{Q})\mathbf{u}_0$, which

gives

$$\mathbf{M}[r_i \mathbf{u}_0] = -i\partial_{Q_i}[e^{i\mathbf{Q}\cdot\mathbf{r}}\mathbf{M}(\mathbf{Q})\mathbf{u}_0]_{\mathbf{Q}=\mathbf{0}} = -i(\partial_i \mathbf{M})\mathbf{u}_0, \quad (\text{C2})$$

the term from differentiating the exponential being removed by $\mathbf{M}(\mathbf{0})\mathbf{u}_0 = \mathbf{0}$. Hence $\mathbf{B}_i = -2i(\partial_i \mathbf{M})\mathbf{u}_0$, which is Eq. (34). The explicit factor of i is what makes $\mathbf{B}_i$ obey the same particle-hole condition as a gradient in doubled pairing space, by the corollary $\tau_1(\partial_i \mathbf{M})^* \tau_1 = -\partial_i \mathbf{M}$ of Eq. (22).

We next relate the frozen-pairing curvature to the direct Goldstone-sector curvature. Since the strictly linear phase $\phi(\mathbf{r}) = 2\mathbf{q} \cdot \mathbf{r}$ is not itself a periodic fluctuation, we first extract the long-wavelength gradient coefficient using a small periodic phase field, for which the Gaussian expansion is justified, and only then evaluate it for the uniform twist. Consider $\Phi(\mathbf{r}) = e^{i\phi(\mathbf{r})\tau_3}\Psi_0$ with $\phi(\mathbf{r}_{lS}) = \frac{1}{\sqrt{N_c}}\sum_{\mathbf{Q}} e^{i\mathbf{Q}\cdot\mathbf{r}_{lS}}\phi(\mathbf{Q})$. For small ϕ , $\delta\Phi(\mathbf{r}) = \phi(\mathbf{r})\mathbf{u}_0 - \phi^2(\mathbf{r})\Psi_0/2 + \dots$, where the ϕ^2 term multiplies the first variation of the action and therefore does not contribute at the saddle. To quadratic order in ϕ , Eq. (19) gives $\mathcal{S}_{\text{eff}}^{(2)} = \frac{\beta}{2}\sum_{\mathbf{Q}} |\phi(\mathbf{Q})|^2 \langle \mathbf{u}_0 | \mathbf{M}(\mathbf{Q}) | \mathbf{u}_0 \rangle$. Expanding about $\mathbf{Q} = \mathbf{0}$ gives $\langle \mathbf{u}_0 | \mathbf{M}(\mathbf{Q}) | \mathbf{u}_0 \rangle = \frac{1}{2}\sum_{ij} Q_i Q_j \langle \mathbf{u}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{u}_0 \rangle + O(Q^4)$, the zeroth-order term vanishing by the Goldstone condition and the linear term by the evenness of $\langle \mathbf{u}_0 | \mathbf{M}(\mathbf{Q}) | \mathbf{u}_0 \rangle$ in $\mathbf{Q}$ (Sec. III C). Therefore $\mathcal{S}_{\text{eff}}^{(2)} = \frac{\beta}{4}\sum_{ij} \langle \mathbf{u}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{u}_0 \rangle \sum_{\mathbf{Q}} Q_i Q_j |\phi(\mathbf{Q})|^2$. For the periodic test phase, Parseval's identity in the above convention gives $\sum_{\mathbf{Q}} Q_i Q_j |\phi(\mathbf{Q})|^2 = \sum_l [\nabla_i \phi(\mathbf{r}_{lS})][\nabla_j \phi(\mathbf{r}_{lS})]$ for any fixed orbital S . Using Eq. (C1) and $\mathcal{S}_{\text{eff}} = \beta N_c \Omega_0$ for a static configuration,

$$\delta\Omega_0|_{\Psi=\Psi_0} = \frac{|\Delta_0|^2}{2N_c} \sum_{ij} \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle \sum_l [\nabla_i \phi(\mathbf{r}_{lS})][\nabla_j \phi(\mathbf{r}_{lS})]. \quad (\text{C3})$$

Having extracted the local gradient coefficient in Eq. (C3), we evaluate it for the uniform twist, $\nabla_i \phi = 2q_i$, for which the cell sum gives a factor N_c and $\delta\Omega_0|_{\Psi=\Psi_0} = 2|\Delta_0|^2 \sum_{ij} q_i q_j \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle$. Comparing with the quadratic expansion $\delta\Omega_0 = \frac{1}{2}\sum_{ij} (\partial^2 \Omega_0 / \partial q_i \partial q_j) q_i q_j$ yields

$$\left. \frac{\partial^2 \Omega_0}{\partial q_i \partial q_j} \right|_{\mathbf{q}=\mathbf{0}, \Psi=\Psi_0} = 4|\Delta_0|^2 \langle \mathbf{w}_0 | \partial_i \partial_j \mathbf{M} | \mathbf{w}_0 \rangle, \quad (\text{C4})$$

which is Eq. (38) after multiplication by N_c . Equation (37) then follows by projecting Eq. (11) onto the massive eigenvectors of $\mathcal{H}(\mathbf{0}) = \mathbf{M}(\mathbf{0})$ and using Eqs. (34), (C1) and (29), as done in Sec. IV.

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